

MODULE 21 | MASTER GENERAL FITNESS COACHING RULES

305 coaching rules across 20 subsections. Format: [Rxxx] IF condition THEN action.

21.1 Assessment Rules

[R001] IF person has never completed a fitness assessment → **THEN** run full Module 2 baseline assessment before assigning any program

[R002] IF person reports chest pain or severe shortness of breath at rest → **THEN** require medical clearance before any exercise prescription

[R003] IF person is pregnant or postpartum → **THEN** require medical clearance and route to Module 15 conservative guidance

[R004] IF person has BMI below 18.5 → **THEN** flag as underweight and route to Module 15 underweight protocol

[R005] IF person has BMI 25-29.9 → **THEN** flag as overweight and route to Module 15 overweight protocol, low-impact start

[R006] IF person has BMI 30+ → **THEN** flag as obese and require extra caution on impact/joint load, route to Module 15

[R007] IF person's daily steps are under 5,000 → **THEN** classify as sedentary and start with Module 6 Program A

[R008] IF person's resting heart rate exceeds 100 bpm at rest with no known cause → **THEN** recommend medical evaluation before intense training

[R009] IF person fails the overhead reach test (cannot raise arms vertical without back arch) → **THEN** prioritize shoulder mobility work before loading overhead presses

[R010] IF person cannot complete a bodyweight squat with heels flat → **THEN** prioritize ankle and hip mobility before squat loading

[R011] IF person reports pain during basic daily movement (walking, sitting, standing) → **THEN** require professional evaluation before structured exercise

[R012] IF person is 65+ and new to exercise → **THEN** begin with low-impact, balance-focused programming

[R013] IF assessment data is incomplete → **THEN** use available data conservatively and default to beginner-level programming

[R014] IF person's energy rating is below 4/10 for 2+ weeks → **THEN** investigate sleep, stress, hydration before assigning new training load

[R015] IF reassessment is due (4-8 weeks since last) → **THEN** trigger Module 2 re-test and update program accordingly

[R016] IF person reports no known health conditions and moderate baseline fitness → **THEN** proceed directly to standard program assignment

[R017] IF waist-to-height ratio exceeds 0.5 → **THEN** flag for general health awareness, no special exercise restriction required

[R018] IF single-leg stand time is under 10 seconds → **THEN** add dedicated balance training to the program immediately

[R019] IF person cannot recall any assessment metrics → **THEN** administer the quick self-check (Module 1.10) as a fast substitute

[R020] IF person provides conflicting self-reported fitness level and test results → **THEN** trust the test results over self-report for program calibration

[R276] IF person's mobility assessment shows restriction in multiple joints simultaneously → **THEN** prioritize the most functionally limiting joint first rather than addressing all at once

[R277] IF person requests reassessment earlier than the 4-8 week window → **THEN** allow it but caution that short-interval data may show more noise than signal

21.2 Beginner Coaching Rules

[R021] IF person has never exercised before → **THEN** assign Module 6 Program A (Absolute Beginner)

[R022] IF beginner reports high anxiety about starting → **THEN** start with home-only, zero-equipment sessions for the first 2 weeks

[R023] IF beginner completes 4 consecutive weeks with good adherence → **THEN** progress to Module 7 intermediate programming

[R024] IF beginner misses more than 50% of planned sessions in week 1 → **THEN** reduce program volume by half rather than abandoning entirely

[R025] IF beginner reports severe soreness after first session → **THEN** reassure as normal, recommend active recovery, do not skip next session entirely

[R026] IF beginner cannot complete a single full push-up → **THEN** regress to wall or incline push-up variation immediately

[R027] IF beginner cannot hold a plank for 10 seconds → **THEN** regress to wall plank or knee plank

[R028] IF beginner expresses confusion about exercise terminology → **THEN** use plain-language cues instead of technical terms

[R029] IF beginner asks for a faster path to visible results → **THEN** reset expectations to the 4-8 week typical adaptation timeline

[R030] IF beginner has access to a gym but no experience using equipment → **THEN** start with bodyweight and dumbbells only for the first 2-3 weeks

[R031] IF beginner reports feeling judged or self-conscious → **THEN** recommend home training or off-peak gym hours

[R032] IF beginner is returning after years of inactivity → **THEN** treat identically to a true beginner regardless of past fitness history

[R033] IF beginner completes all sets/reps with no fatigue at all → **THEN** increase reps or progress exercise difficulty next session

[R034] IF beginner reports dizziness during a session → **THEN** stop the session immediately and recommend hydration and rest

[R035] IF beginner asks how many days per week to train → **THEN** recommend 3 non-consecutive days as the default starting frequency

[R036] IF beginner has very limited available equipment → **THEN** default to bodyweight-only programming from Module 6 Program D

[R037] IF beginner reports boredom within the first 2 weeks → **THEN** introduce one new exercise variation while keeping core structure

[R038] IF beginner asks about supplements before establishing consistency → **THEN** redirect focus to consistency and basic nutrition first

[R039] IF beginner's form breaks down before reaching target reps → **THEN** reduce reps to the last point of good form, not the prescribed number

[R040] IF beginner reports a previous injury with no current symptoms → **THEN** proceed with standard programming but monitor that movement pattern closely

[R278] IF beginner asks whether to join a gym immediately → **THEN** recommend building home consistency first before adding a gym commitment

[R279] IF beginner reports liking one exercise far more than others → **THEN** allow more frequent use of the preferred exercise while still covering all patterns weekly

21.3 Cardio Coaching Rules

[R041] IF person has knee or joint sensitivity → **THEN** prioritize walking, cycling, or swimming over running or jogging

[R042] IF person has zero equipment and no gym access → **THEN** prioritize walking, jogging, stair climbing, or skipping rope

[R043] IF person is a complete beginner to structured cardio → **THEN** start with brisk walking before introducing jogging or running

[R044] IF person reports boredom with current cardio routine → **THEN** introduce cross-training across 2-3 different cardio activities per week

[R045] IF person has under 15 minutes available for cardio → **THEN** prioritize skipping rope or stair climbing for time efficiency

[R046] IF person's goal is general fitness rather than performance → **THEN** target roughly 80% of cardio time at light-moderate intensity, 20% vigorous

[R047] IF person cannot sustain a conversation during cardio (fails the talk test) → **THEN** reduce intensity to moderate zone immediately

[R048] IF person reports chronic shin pain during running → **THEN** substitute cycling or swimming until pain resolves

[R049] IF person wants to progress from walking to jogging → **THEN** use run-walk intervals (1 min jog, 2 min walk) before continuous jogging

[R050] IF person trains cardio every single day at high intensity → **THEN** insert at least one low-intensity or rest day per week

[R051] IF person's cardio sessions have plateaued for 4+ weeks → **THEN** introduce interval training or increase pace/incline

[R052] IF person reports hot, humid climate conditions → **THEN** shift cardio sessions to early morning or evening and increase hydration

[R053] IF person is training for general health, not a race → **THEN** do not exceed 5-6 cardio sessions per week without a rest day

[R054] IF person experiences ankle/foot pain while walking or jogging → **THEN** check footwear first before reducing activity

[R055] IF person wants the most time-efficient cardio option → **THEN** recommend skipping rope intervals

[R056] IF person has access to a pool → **THEN** offer swimming as a near-zero-impact cardio option

[R057] IF person reports feeling unmotivated for steady-state cardio → **THEN** switch to interval format for variety and engagement

[R058] IF person is recovering from a lower-body injury → **THEN** default to swimming or cycling until cleared for impact activity

[R059] IF person asks what intensity to train at without a heart rate monitor → **THEN** use the talk test as the default intensity gauge

[R060] IF person reports cardio sessions feel easier than 4 weeks ago at the same pace → **THEN** confirm this as a positive fitness improvement signal and progress intensity

[R280] IF person asks whether walking counts as real exercise → **THEN** confirm walking is a legitimate, evidence-supported cardio activity

[R281] IF person wants to combine walking and strength in one efficient session → **THEN** use a walk-to-warm-up followed by a bodyweight circuit format

21.4 Strength Training Rules

[R061] IF person is a complete beginner to strength training → **THEN** prioritize bodyweight versions of all six fundamental patterns for 2-4 weeks

[R062] IF person completes 15+ reps of an exercise with good form → **THEN** progress to a harder variation or add resistance

[R063] IF person reports strength imbalance between sides → **THEN** emphasize unilateral exercises such as lunges and single-arm rows

[R064] IF person has limited equipment for strength training → **THEN** build the program from bodyweight push/pull/squat/lunge/hinge/core exercises

[R065] IF person trains strength 2 days per week → **THEN** use full-body sessions covering all movement patterns each session

[R066] IF person trains strength 4+ days per week → **THEN** consider an upper-lower split for better recovery distribution

[R067] IF person reports pain (not fatigue) during a specific pattern → **THEN** substitute a regression variation for that pattern immediately

[R068] IF person has not increased reps, sets, or load in 4+ weeks → **THEN** apply progressive overload via one variable change

[R069] IF person's push volume far exceeds pull volume → **THEN** rebalance the program toward more pulling exercises

[R070] IF person cannot maintain neutral spine during a hinge movement → **THEN** regress to glute bridge until hinge mechanics improve

[R071] IF person wants to train strength every single day → **THEN** cap consecutive strength days at 2 and require at least 1 rest day per week

[R072] IF person reports excessive soreness lasting beyond 5 days → **THEN** reduce volume or intensity for that muscle group next session

[R073] IF person has access to a full gym for the first time → **THEN** introduce machines gradually alongside familiar bodyweight movements

[R074] IF person prioritizes functional, real-world strength → **THEN** emphasize carries, compound lifts, and multi-planar movement

[R075] IF person is unsure how much weight to use for a new exercise → **THEN** select a load where the last 2-3 reps of the target range feel challenging

[R076] IF person reports a session felt too easy with reps still in reserve → **THEN** increase load or progress to a harder variation next session

[R077] IF person reports significant form breakdown before completing planned reps → **THEN** the session intensity was too high; reduce load next time

[R078] IF person wants to build strength with minimal time per session → **THEN** prioritize compound exercises over isolation work

[R079] IF person has a desk job and reports tight hip flexors limiting hinge depth → **THEN** pair hinge training with daily hip flexor mobility work

[R080] IF person is uncertain whether to train to failure → **THEN** recommend stopping 1-2 reps short of failure for general fitness goals

[R282] IF person asks if bodyweight training alone is enough long-term → **THEN** confirm it remains effective long-term with progressive variation and tempo changes

[R283] IF person wants to train strength but has chronic wrist discomfort → **THEN** substitute forearm-based or machine-based pressing variations

21.5 Mobility Coaching Rules

[R081] IF person sits 6+ hours per day → **THEN** prioritize daily hip flexor, thoracic spine, and neck mobility work

[R082] IF person has limited shoulder overhead range → **THEN** prioritize wall slides and doorway chest stretch before overhead pressing

[R083] IF person is warming up before a workout → **THEN** use dynamic stretching only, not static holds

[R084] IF person is cooling down after a workout → **THEN** static stretching is appropriate while muscles are warm

[R085] IF person's squat depth is limited in assessment → **THEN** prioritize ankle dorsiflexion and hip mobility before adding squat load

[R086] IF person has under 5 minutes available for mobility → **THEN** assign the 3-minute travel/office mobility routine

[R087] IF a specific joint is flagged restricted in assessment → **THEN** prioritize that joint's dedicated mobility work twice daily

[R088] IF mobility work causes sharp pain rather than mild stretch tension → **THEN** stop immediately and recommend professional evaluation

[R089] IF person reports general stiffness with no specific painful area → **THEN** assign the full-body 10-minute daily mobility routine

[R090] IF person is a frequent traveler with inconsistent schedule → **THEN** assign the compressed 3-minute mobility routine for travel days

[R091] IF person wants to improve flexibility specifically, not just mobility → **THEN** add dedicated static stretching sessions 3-5x per week, 20-30 second holds

[R092] IF person reports tight hamstrings from prolonged sitting → **THEN** address hip flexor tightness alongside hamstring stretching

[R093] IF person cannot perform a deep squat hold without falling backward → **THEN** regress to a supported squat hold using a wall or doorframe

[R094] IF person has restricted thoracic rotation → **THEN** prioritize open book stretch and thread the needle drills

[R095] IF person reports neck stiffness from screen use → **THEN** assign daily chin tucks and neck rotation drills

[R096] IF person's ankle dorsiflexion is limited on the knee-to-wall test → **THEN** prioritize ankle mobility before progressing squat depth or load

[R097] IF person wants a single daily mobility habit that covers the most ground → **THEN** assign the 10-minute full-body routine over isolated single-joint work

[R098] IF person reports stretching causes joint instability or pain beyond mild tension → **THEN** discontinue that stretch and recommend professional evaluation

[R099] IF person is preparing for a loaded lift (e.g., goblet squat) → **THEN** perform exercise-specific dynamic mobility immediately beforehand

[R100] IF person has good mobility scores across all assessment tests → **THEN** maintain current mobility work as upkeep rather than increasing volume

[R284] IF person asks how long mobility improvements take to appear → **THEN** set expectations of initial feel improvement in 1-2 weeks, lasting range changes in 4-8 weeks

21.6 Recovery Coaching Rules

[R101] IF resting heart rate is elevated 5+ bpm above baseline for 3+ consecutive days → **THEN** recommend an active recovery or full rest day

[R102] IF soreness from a session lasts beyond 4-5 days → **THEN** reduce volume or intensity for that muscle group next session

[R103] IF person reports high life or work stress alongside training fatigue → **THEN** prioritize rest and stress-management practices over training volume

[R104] IF person has trained intensely for 6-10 weeks without a lighter week → **THEN** recommend a deload week

[R105] IF person is mildly ill with above-the-neck symptoms only → **THEN** light activity is generally acceptable

[R106] IF person has below-the-neck symptoms or fever → **THEN** recommend rest until fully recovered before resuming training

[R107] IF person feels generally fatigued but not exhausted → **THEN** assign active recovery (light walk, gentle mobility)

[R108] IF person feels significantly run down with declining performance → **THEN** assign a complete passive rest day

[R109] IF person trains 5+ days per week → **THEN** ensure at least 1 full passive rest day is scheduled

[R110] IF person is returning from illness or a multi-week break → **THEN** resume at reduced volume and intensity, prioritizing consistency

[R111] IF person reports persistent fatigue despite adequate training volume → **THEN** review sleep quality and duration before adjusting training

[R112] IF person asks about post-workout nutrition timing → **THEN** explain that a balanced meal within a few hours of training is sufficient

[R113] IF person trains the same muscle group on consecutive days → **THEN** recommend at least 48 hours between sessions targeting that muscle group

[R114] IF person reports symptoms consistent with overtraining (mood decline, poor sleep, flat performance) → **THEN** recommend an extended deload or rest period

[R115] IF person asks whether stretching alone is sufficient recovery → **THEN** clarify that sleep, nutrition, and rest days matter more than stretching alone

[R285] IF person asks if recovery tools like massage guns are necessary → **THEN** clarify they are optional supplements, not substitutes for sleep and nutrition

21.7 Sleep Coaching Rules

[R116] IF person reports under 6 hours of sleep regularly → **THEN** flag sleep as the priority recovery lever before increasing training load

[R117] IF person trains intensely in the evening and reports trouble falling asleep → **THEN** recommend shifting intense training earlier or allowing 1-2 hours wind-down

[R118] IF person has inconsistent sleep and wake times → **THEN** recommend a consistent schedule even on rest days

[R119] IF person uses screens within 30 minutes of bedtime → **THEN** recommend reducing screen exposure before sleep

[R120] IF person consumes caffeine in the afternoon and reports poor sleep → **THEN** recommend avoiding caffeine after early afternoon

[R121] IF person is a shift worker with irregular sleep hours → **THEN** anchor sleep environment consistency to personal schedule, not clock time

[R122] IF person reports waking up still tired despite 7+ hours in bed → **THEN** investigate sleep quality factors (environment, screens, stress) rather than just duration

[R123] IF person asks how much sleep is needed for recovery → **THEN** recommend 7-9 hours nightly for most adults

[R124] IF person reports napping disrupts nighttime sleep → **THEN** recommend limiting naps to 20-30 minutes earlier in the day

[R125] IF person's sleep has declined over several weeks alongside a training plateau → **THEN** treat sleep as the likely primary cause before changing the training program

[R286] IF person asks if weekend sleep-ins can compensate for weekday sleep debt → **THEN** clarify partial compensation is possible but consistent nightly sleep is preferable

21.8 Hydration Coaching Rules

[R126] IF person reports dark urine color → **THEN** recommend increasing water intake throughout the day

[R127] IF person trains intensely or in hot conditions → **THEN** recommend an additional 500ml-1L of water beyond baseline intake

[R128] IF person asks if coffee or tea counts toward hydration → **THEN** confirm both contribute positively to daily fluid intake

[R129] IF person reports muscle cramping during exercise → **THEN** review hydration and electrolyte intake as one contributing factor

[R130] IF person trains under 60 minutes at moderate intensity → **THEN** water alone is typically sufficient without electrolyte drinks

[R131] IF person trains over 90 minutes or in heavy heat/humidity → **THEN** recommend electrolyte replacement alongside water

[R132] IF person reports excessive thirst and fatigue during a session → **THEN** treat as a dehydration signal and increase fluid intake immediately

[R133] IF person asks about a simple daily hydration habit → **THEN** recommend starting the day with a glass of water and keeping a bottle visible

[R134] IF person reports drinking large volumes of water very quickly during exercise → **THEN** recommend smaller, more frequent sips instead

[R135] IF person has no clear hydration habit at all → **THEN** set pale yellow urine color as the simple daily target indicator

[R287] IF person asks if overhydration is a real concern for general fitness training → **THEN** clarify it is rare and mainly relevant to prolonged endurance events

21.9 Habit Formation Rules

[R136] IF person is starting a new fitness habit → **THEN** begin with a version small enough to complete even on a bad day

[R137] IF person reports inconsistent follow-through → **THEN** apply habit stacking to an existing daily routine

[R138] IF person misses a scheduled workout → **THEN** resume at the next opportunity without trying to make up missed sessions

[R139] IF person misses two consecutive days → **THEN** flag as a higher-risk point for habit loss and prioritize immediate return

[R140] IF person reports low motivation but a stable schedule → **THEN** offer a reduced minimum viable workout rather than skipping entirely

[R141] IF person has sustained a habit for 2-3+ months consistently → **THEN** shift focus toward progressive program advancement

[R142] IF person is in the first 1-3 weeks of a new habit → **THEN** treat as the initiation phase and minimize complexity and decision-making

[R143] IF person is in weeks 4-10 of a new habit → **THEN** treat as the learning phase and allow gradual increases in structure

[R144] IF person has trained consistently for 10+ weeks → **THEN** treat as the stability phase and allow more program autonomy

[R145] IF person asks how to build identity around fitness → **THEN** reframe goals around becoming someone who shows up consistently, not just outcomes

[R146] IF person reports a broken streak after a long consistent stretch → **THEN** resume at reduced intensity and avoid all-or-nothing thinking

[R147] IF person has no existing daily routine to stack a habit onto → **THEN** identify the most fixed daily anchor (waking, a meal) to attach the new habit to

[R148] IF person asks for the single most important habit-building tip → **THEN** recommend making the first version of the habit almost too easy to skip

[R149] IF person reports habit feels effortful even after months → **THEN** reassure this is normal for some people and does not indicate failure

[R150] IF person wants to add a second fitness habit while still building the first → **THEN** delay the second habit until the first reaches the stability phase

[R288] IF person wants to build two habits simultaneously from day one → **THEN** recommend sequencing them instead, starting with the highest-priority habit

21.10 Motivation Rules

[R151] IF person reports low motivation for an extended period → **THEN** investigate underlying connection to deeper personal reasons for training

[R152] IF person's motivation is tied only to appearance goals → **THEN** introduce process-based and identity-based framing alongside appearance goals

[R153] IF person reports comparing themselves negatively to others → **THEN** redirect focus to personal trend and starting-point comparison only

[R154] IF person hits a small milestone such as a first full push-up → **THEN** acknowledge it specifically and concretely rather than generically

[R155] IF person reports exercise feels like a chore with no enjoyment → **THEN** introduce variety or a new activity format to rebuild engagement

[R156] IF person lacks a workout partner or accountability system → **THEN** suggest a remote check-in partner or app-based accountability

[R157] IF person's motivation spikes after watching fitness content but fades quickly → **THEN** redirect toward sustainable small actions rather than inspired bursts

[R158] IF person reports motivation dips during exam or high-stress periods → **THEN** reduce volume but maintain minimal consistency rather than full stop

[R159] IF person asks how to stay motivated without external rewards → **THEN** connect training to intrinsic values like energy, health, and capability

[R160] IF person reports frustration with slow visible progress → **THEN** redirect attention to non-scale wins such as energy, strength, and mood

[R289] IF person responds well to gamification and streak tracking → **THEN** lean into visible progress tracking as a core motivational tool for this person

21.11 Consistency Rules

[R161] IF person keeps skipping scheduled workouts → **THEN** reassess if the scheduled time realistically fits their life and adjust

[R162] IF person returns after a multi-week break → **THEN** resume at roughly 50-60% of previous volume and intensity

[R163] IF person is consistent but reports no progress → **THEN** verify true consistency over 4-6 weeks before assuming a plateau

[R164] IF person's schedule changes frequently week to week → **THEN** build a flexible weekly target instead of fixed daily times

[R165] IF person loses interest in their current routine → **THEN** introduce variety in exercises, format, or activity type

[R166] IF person experiences illness or injury interrupting consistency → **THEN** focus on what is still safely possible rather than full stoppage

[R167] IF person's support system does not support their fitness goals → **THEN** seek external accountability via community or app-based support

[R168] IF person reports a particularly stressful or busy life period → **THEN** reduce the routine to a minimum viable version rather than stopping entirely

[R169] IF person asks if consistency fluctuation over years is normal → **THEN** confirm this is expected and emphasize systems for returning quickly

[R170] IF person wants to track consistency simply → **THEN** recommend a visual streak tracker or simple Y/N daily log

[R171] IF person has completed 1 month with perfect adherence → **THEN** introduce a small new challenge or progression to sustain engagement

[R172] IF person reports two consecutive missed sessions → **THEN** treat this as the highest-risk point for habit loss and re-engage immediately

[R173] IF person asks whether to make up a missed workout → **THEN** advise resuming the normal schedule rather than doubling up

[R174] IF person's consistency is high but enjoyment is low → **THEN** introduce one enjoyable activity per week alongside structured training

[R175] IF person reports family or social obligations are conflicting with fixed workout times → **THEN** shift to a flexible weekly-total approach instead of fixed days

[R290] IF person asks how to rebuild consistency after a guilt-driven break → **THEN** reframe the break as normal and focus only on the next session, not the gap

21.12 Plateau Rules

[R176] IF person reports no progress for 4-6+ weeks despite consistency → **THEN** treat as a genuine plateau requiring a deliberate program adjustment

[R177] IF strength has plateaued → **THEN** introduce exercise variation, adjust rep ranges, or apply a deload then resume overload

[R178] IF cardio fitness has plateaued → **THEN** restructure intensity distribution by adding intervals or more low-intensity volume

[R179] IF mobility or flexibility has plateaued → **THEN** increase stretching frequency or hold duration before assuming a hard ceiling

[R180] IF person assumes a plateau but has been inconsistent → **THEN** address consistency first before treating it as a true plateau

[R181] IF person assumes a plateau but recovery factors have declined → **THEN** address sleep, stress, and nutrition before changing the training program

[R182] IF person has trained the exact same program unchanged for 8+ weeks → **THEN** introduce planned variation to break the adaptation plateau

[R183] IF person hits a plateau shortly after a period of poor sleep → **THEN** treat sleep as the likely primary cause

[R184] IF person reports a plateau despite increasing training volume → **THEN** check whether recovery has kept pace with the added volume

[R185] IF person wants the fastest way to break a strength plateau → **THEN** recommend a deload week followed by fresh progressive overload

[R186] IF person wants the fastest way to break a cardio plateau → **THEN** recommend adding structured high-intensity intervals

[R187] IF person's plateau coincides with high external life stress → **THEN** prioritize stress and recovery management over training changes

[R188] IF person mistakes normal week-to-week variation for a plateau → **THEN** review trend data over 4-6 weeks rather than single sessions

[R189] IF person has plateaued and reports boredom with current exercises → **THEN** rotate to new but related exercise variations

[R190] IF person plateaus after months without ever taking a deload → **THEN** schedule an immediate deload week regardless of program timing

[R291] IF person experiences a plateau in balance or coordination specifically → **THEN** increase the difficulty of balance drills via reduced base of support or eyes-closed variations

21.13 Time-Constrained User Rules

[R191] IF person has under 20 minutes available per session → **THEN** prioritize a full-body circuit of compound exercises

[R192] IF person has under 10 minutes available → **THEN** assign a 3-4 round bodyweight circuit of squats, push-ups, and plank

[R193] IF person consistently runs out of time mid-workout → **THEN** pre-select a shorter fallback version of the session in advance

[R194] IF person wants maximum benefit per minute trained → **THEN** prioritize compound, multi-joint exercises over isolation work

[R195] IF person can only train 2 days per week due to time → **THEN** use full-body sessions covering all movement patterns each session

[R196] IF person asks whether short workouts are actually effective → **THEN** confirm that consistent short sessions provide meaningful benefit

[R197] IF person has variable daily time availability → **THEN** provide a menu of 10, 20, and 30-minute session options

[R198] IF person has zero time on a given day → **THEN** offer a single-set minimum viable workout rather than skipping entirely

[R199] IF person wants to combine strength and cardio in minimal time → **THEN** use a circuit format alternating strength and cardio movements

[R200] IF person asks if supersets save time without sacrificing results → **THEN** confirm supersets maintain stimulus while reducing total session time

[R292] IF person wants a true 5-minute emergency session → **THEN** prescribe 3 rounds of 10 squats, 10 push-ups, and a 20 second plank with no rest

21.14 Student Lifestyle Rules

[R201] IF person is a student with a packed academic schedule → **THEN** assign short 20-30 minute bodyweight sessions fitting between classes

[R202] IF student reports an upcoming exam period → **THEN** reduce session count to 2-3 short sessions and treat exercise as stress relief

[R203] IF student lives in a hostel with no equipment → **THEN** assign a zero-equipment bodyweight circuit program

[R204] IF student has access to campus walking paths → **THEN** use campus walking as built-in daily cardio activity

[R205] IF student reports a tight budget → **THEN** default entirely to bodyweight training with no equipment cost

[R206] IF student is going home for semester break → **THEN** switch temporarily to the Traveler lifestyle guidance

[R207] IF student reports irregular class schedule day to day → **THEN** use a flexible weekly target rather than fixed daily session times

[R208] IF student shares a small dorm room with roommates → **THEN** select stationary, space-efficient exercises over those needing room to move

[R209] IF student asks how to stay consistent despite social distractions → **THEN** anchor workouts to a fixed daily routine cue, such as after waking

[R210] IF student reports low privacy for exercising → **THEN** schedule sessions during quieter hours or use only standing exercises

[R293] IF student reports inconsistent meal timing affecting training energy → **THEN** recommend training at the most reliably fed point in their daily schedule

21.15 Office Worker Rules

[R211] IF person works a desk job 8+ hours per day → **THEN** assign movement breaks every 30-60 minutes alongside structured training

[R212] IF office worker reports neck and shoulder tightness → **THEN** prioritize daily desk mobility routine targeting hips, chest, and thoracic spine

[R213] IF office worker has a lunch break available → **THEN** recommend using it for a brisk walk or short workout

[R214] IF office worker has access to a standing desk → **THEN** recommend alternating sitting and standing throughout the day

[R215] IF office worker reports no time before or after work → **THEN** default to lunch-break or movement-break sessions

[R216] IF office worker sits in back-to-back meetings most of the day → **THEN** recommend walking during phone calls when possible

[R217] IF office worker reports posture discomfort by end of day → **THEN** assign the office ergonomic self-checklist alongside posture-corrective exercises

[R218] IF office worker commutes by car with no walking built in → **THEN** recommend parking further away or taking a walking detour

[R219] IF office worker has access to building stairs → **THEN** recommend taking stairs instead of elevators as default daily movement

[R220] IF office worker reports persistent low energy in the afternoon → **THEN** recommend a brief walk during the afternoon slump instead of caffeine alone

[R294] IF office worker reports wrist strain from typing affecting push exercises → **THEN** substitute fist or forearm push-up variations temporarily

21.16 Frequent Traveler Rules

[R221] IF person travels frequently for work → **THEN** default to a bodyweight circuit usable in any hotel room

[R222] IF person is on a travel or transit day → **THEN** prioritize walking over a structured workout

[R223] IF person has hotel gym access → **THEN** use a modified version of their normal program with available equipment

[R224] IF person reports jet lag affecting training capacity → **THEN** allow 1-2 days reduced intensity before resuming normal training

[R225] IF person is traveling for multiple weeks → **THEN** accept reduced training volume as a reasonable trade-off, not a failure

[R226] IF person asks what equipment is worth packing while traveling → **THEN** recommend a resistance band as the most practical travel equipment

[R227] IF person has returned home after travel → **THEN** resume normal programming promptly rather than easing back in gradually

[R228] IF person is in an unfamiliar hotel gym with unfamiliar equipment → **THEN** stick to familiar movement patterns rather than experimenting

[R229] IF person reports inconsistent meal and sleep timing while traveling → **THEN** prioritize sleep consistency over strict workout adherence during travel

[R230] IF person combines frequent travel with a desk job between trips → **THEN** alternate between the office worker routine at home and traveler routine on trips

[R295] IF person travels across multiple time zones frequently → **THEN** prioritize sleep schedule stabilization over training schedule in the first 1-2 days

21.17 Older Adult Rules

[R231] IF person is 65+ and new to exercise → **THEN** begin with low-impact, balance-focused programming

[R232] IF older adult reports joint sensitivity → **THEN** prioritize swimming, cycling, and walking over high-impact cardio

[R233] IF older adult has poor single-leg stand results → **THEN** prioritize dedicated balance training immediately

[R234] IF older adult has an underlying health condition → **THEN** require medical clearance before starting or changing a program

[R235] IF older adult asks if strength training is safe → **THEN** confirm it is safe and beneficial when appropriately progressed

[R236] IF older adult reports difficulty rising from a chair unassisted → **THEN** prioritize the sit-to-stand functional pattern

[R237] IF older adult reports difficulty getting up from the floor → **THEN** prioritize the floor get-up functional pattern

[R238] IF older adult is beginning resistance training for the first time → **THEN** use light loads with a focus on full range of motion and control

[R239] IF older adult reports fear of falling → **THEN** prioritize balance and lower-body strength training together

[R240] IF older adult is recovering from a recent fall → **THEN** require medical clearance and use conservative, supported exercise selection

[R296] IF older adult reports good baseline fitness for their age group → **THEN** still include balance and functional training as preventive maintenance

21.18 Overweight Population Rules

[R241] IF person has BMI 25-29.9 and is new to exercise → **THEN** start with low-impact activities such as walking, swimming, and cycling

[R242] IF overweight beginner reports joint discomfort during standard squats → **THEN** regress to box squats or reduced range of motion

[R243] IF overweight beginner cannot complete a standard push-up → **THEN** regress to an incline or wall push-up

[R244] IF overweight beginner asks about training frequency → **THEN** recommend 2-3 sessions per week of 20-30 minutes as a sustainable start

[R245] IF overweight beginner reports slower visible progress than expected → **THEN** redirect focus to non-scale victories such as energy and strength

[R246] IF overweight beginner asks if high-impact exercise should be avoided forever → **THEN** clarify that impact tolerance can be built gradually over time

[R247] IF overweight beginner reports joint pain with running specifically → **THEN** substitute walking, cycling, or swimming instead

[R248] IF overweight beginner is ready to progress after 4+ weeks of consistency → **THEN** gradually increase session duration and introduce modest impact activities

[R249] IF overweight individual asks about strength training safety → **THEN** confirm it is safe and beneficial when modified appropriately

[R250] IF overweight beginner reports discouragement about pace of change → **THEN** emphasize that consistency over months produces meaningful results

[R297] IF overweight individual asks about swimming as a primary activity → **THEN** confirm it is an excellent low-impact option for this profile

21.19 Underweight Population Rules

[R251] IF person has BMI below 18.5 → **THEN** prioritize strength training over excessive cardio volume

[R252] IF underweight person asks about cardio volume → **THEN** caution against excessive cardio that could increase calorie expenditure further

[R253] IF underweight beginner is starting strength training → **THEN** use a full-body routine 3x per week with compound bodyweight or light-resistance exercises

[R254] IF underweight person reports low energy during sessions → **THEN** recommend reviewing adequacy of calorie and nutrient intake alongside training

[R255] IF underweight person asks if bodyweight training alone can build muscle → **THEN** confirm it can, especially in early training stages

[R256] IF underweight person is progressing well in strength but plateaus in muscle gain → **THEN** recommend adding external resistance once bodyweight training is mastered

[R257] IF underweight individual reports frequent fatigue between sessions → **THEN** prioritize recovery and nutrition support over increasing training volume

[R258] IF underweight person asks whether to prioritize strength or cardio → **THEN** recommend prioritizing strength training for this goal profile

[R259] IF underweight individual shows signs of overtraining without adequate fueling → **THEN** reduce training volume and flag nutrition as a priority

[R260] IF underweight person has significant body composition concerns alongside symptoms → **THEN** recommend medical consultation before continuing self-directed programming

[R298] IF underweight person reports feeling intimidated by gym strength equipment → **THEN** start with home bodyweight and band training to build confidence first

21.20 Program Progression Rules

[R261] IF beginner completes 4 consecutive weeks with good adherence → **THEN** progress to intermediate-level programming

[R262] IF intermediate plateaus on their assigned program for 6+ weeks → **THEN** rotate to a different intermediate program variant

[R263] IF person completes the top rep or time target for a progression step across 2 sessions → **THEN** advance to the next step in that progression tree

[R264] IF person cannot complete the minimum target for their current progression step → **THEN** remain at or drop to the prior step

[R265] IF person reports pain at a specific progression step → **THEN** drop to the prior step and review movement quality

[R266] IF a program cycle of 4, 8, or 12 weeks completes → **THEN** reassess baseline fitness before designing the next cycle

[R267] IF person has trained the same program unchanged for 8+ weeks with good progress → **THEN** introduce planned variation to sustain continued adaptation

[R268] IF person's training age is low regardless of chronological age → **THEN** expect faster initial progress and use simpler programming

[R269] IF person's training age is high → **THEN** use more precisely targeted, progressively overloaded programming

[R270] IF person wants to know when to progress an exercise's difficulty → **THEN** use 15+ reps with good form as the general progression trigger

[R271] IF person's goals shift mid-program → **THEN** reassess with the goal differentiation framework and adjust FITT variables

[R272] IF person asks how often a program should be updated → **THEN** recommend every 4-8 weeks to reassess and introduce new variation

[R273] IF person is uncertain whether to progress a beginner program → **THEN** use the four-week minimum adherence threshold before progressing

[R274] IF person reports a program feels stagnant despite progression markers being met → **THEN** introduce a different program format, not just harder exercises

[R275] IF person achieves all benchmarks in a program ahead of schedule → **THEN** progress early rather than waiting for the full planned duration

[R299] IF person asks whether progression should ever skip a step → **THEN** advise against skipping steps even when eager, to protect form and safety

[R300] IF person reaches the final step of a progression tree → **THEN** transition to load-based or volume-based progression beyond the tree

[R301] IF person regresses temporarily due to illness or a break → **THEN** treat the regression as expected and not a true loss of capability

[R302] IF person requests a progression tree for a pattern not yet covered → **THEN** apply the same step logic used in Module 16 trees to the new pattern

[R303] IF person's progression across different trees is uneven, such as strong squat but weak push → **THEN** allow each tree to progress independently at its own pace

[R304] IF person asks how progression trees relate to program design → **THEN** clarify progression trees determine exercise difficulty, programs determine weekly structure

[R305] IF person wants to self-assess readiness to advance a progression step → **THEN** use the two-consecutive-session full-target-met rule as the standard check

MODULE 22 | GENERAL FITNESS TROUBLESHOOTING ENGINE

210 troubleshooting rules across 8 diagnostic subsections. Format: [Txxx] IF condition THEN action.

22.1 Low Energy Troubleshooting

[T001] IF low energy is reported alongside under 6 hours of sleep → **THEN** diagnose sleep as primary cause; prioritize sleep duration before any training change

[T002] IF low energy is reported alongside high reported stress → **THEN** diagnose stress as primary cause; recommend stress-management practices and lighter training

[T003] IF low energy is reported alongside low water intake or dark urine → **THEN** diagnose dehydration as primary cause; increase water intake immediately

[T004] IF low energy is reported alongside inadequate or irregular meals → **THEN** diagnose nutrition as primary cause; recommend regular, balanced meals

[T005] IF low energy is reported alongside recently increased training volume → **THEN** diagnose training load as primary cause; reduce volume or insert a deload

[T006] IF low energy is reported with no clear sleep, stress, hydration, or nutrition cause → **THEN** diagnose accumulated recovery debt; insert a full rest day and reassess in 48 hours

[T007] IF low energy persists despite addressing sleep, stress, hydration, and training load → **THEN** recommend medical evaluation to rule out underlying health causes

[T008] IF low energy is reported only in the morning → **THEN** investigate sleep quality and consistency of wake time first

[T009] IF low energy is reported only in the afternoon → **THEN** investigate meal timing and blood sugar stability around midday

[T010] IF low energy is reported only on training days → **THEN** investigate whether training load currently exceeds recovery capacity

[T011] IF low energy is reported only on rest days → **THEN** this is atypical; investigate whether rest days lack any movement at all and add light activity

[T012] IF low energy coincides with a new medication or supplement → **THEN** recommend the person discuss this specifically with a healthcare provider

[T013] IF low energy coincides with the start of a new intense program → **THEN** treat as a normal adaptation period and monitor for improvement over 1-2 weeks

[T014] IF low energy is chronic over 4+ weeks with no identifiable lifestyle cause → **THEN** recommend a full medical check-up before continuing self-directed troubleshooting

[T015] IF low energy improves immediately after a short walk → **THEN** confirm this as a sign of low daily movement rather than a deeper recovery issue

[T016] IF low energy is reported by someone newly sedentary after being previously active → **THEN** diagnose deconditioning; reintroduce activity gradually as the primary fix

[T017] IF low energy is reported alongside excessive caffeine intake late in the day → **THEN** flag caffeine timing as a likely contributor to poor sleep and resulting fatigue

[T018] IF low energy is reported alongside skipped meals before training → **THEN** recommend a small pre-workout snack to test for improvement

[T019] IF low energy is reported alongside a recent illness → **THEN** allow continued recovery time before resuming normal training intensity

[T020] IF low energy is reported alongside seasonal changes with less daylight → **THEN** note possible seasonal mood and energy effects and recommend daylight exposure where possible

[T021] IF low energy diagnosis is ambiguous after checking all six factors → **THEN** default to the simplest, lowest-risk intervention first: prioritizing sleep

[T022] IF low energy resolves after one full week of prioritized sleep → **THEN** confirm sleep as the root cause and continue monitoring

[T023] IF low energy does not resolve after addressing sleep, stress, hydration, nutrition, and training load → **THEN** escalate to professional medical evaluation

[T024] IF low energy is reported alongside a major recent life change → **THEN** consider psychological and emotional load as a contributing factor alongside physical ones

[T025] IF person wants a fast self-check method for energy issues → **THEN** recommend the Module 2.8 daily 1-10 energy rating logged for one week

[T153] IF low energy is reported alongside a recently changed work shift pattern → **THEN** treat circadian disruption as a likely contributing factor alongside sleep quantity

[T154] IF low energy is reported but step count has dropped sharply → **THEN** treat reduced daily movement itself as a possible contributing cause

[T155] IF low energy is reported alongside frequent skipped breakfasts → **THEN** test adding a balanced breakfast for one week and reassess

[T156] IF low energy is reported in a person who recently quit caffeine entirely → **THEN** note this as a likely temporary withdrawal effect lasting several days

[T157] IF low energy is reported alongside symptoms suggestive of anemia or nutrient deficiency → **THEN** recommend medical evaluation rather than continued self-troubleshooting

22.2 Cardio Plateau Troubleshooting

[T026] IF cardio fitness has not improved in 4-6+ weeks → **THEN** begin cardio plateau decision tree

[T027] IF cardio plateau coincides with identical pace or route every session → **THEN** introduce pace variation, incline, or route changes

[T028] IF cardio plateau coincides with no interval training ever used → **THEN** add one structured interval session per week

[T029] IF cardio plateau coincides with no low-intensity volume at all → **THEN** add easy-pace volume to build aerobic base

[T030] IF cardio plateau coincides with declining sleep quality → **THEN** address sleep before making further cardio program changes

[T031] IF cardio plateau coincides with inconsistent session frequency → **THEN** address consistency before assuming a true physiological plateau

[T032] IF cardio plateau persists despite varied pace, intervals, and good recovery → **THEN** treat as a genuine plateau and apply a deload week, then resume progression

[T033] IF cardio plateau is specific to one activity only, such as running, but not others → **THEN** this points to activity-specific adaptation; introduce cross-training in that modality

[T034] IF cardio plateau is reported by a complete beginner within the first 2 weeks → **THEN** this is too early to be a true plateau; continue current program

[T035] IF cardio plateau is accompanied by reduced motivation for that activity → **THEN** address motivation and enjoyment alongside the physiological program changes

[T036] IF cardio performance has declined rather than plateaued → **THEN** investigate recovery, illness, or overtraining before assuming normal plateau

[T037] IF cardio plateau is reported in resting heart rate specifically, not session performance → **THEN** review whether sleep and stress have changed recently

[T038] IF cardio plateau is reported alongside a recent increase in life stress → **THEN** prioritize stress and recovery management over additional cardio volume

[T039] IF cardio plateau resolves after adding intervals for 2-3 weeks → **THEN** confirm interval addition as the effective fix and continue that structure

[T040] IF cardio plateau resolves after a deload week → **THEN** confirm accumulated fatigue was the root cause

[T041] IF person wants to know how long to try a fix before reassessing → **THEN** recommend a minimum 3-4 week trial before judging effectiveness

[T042] IF cardio plateau is reported with very high weekly training frequency already at 6+ days → **THEN** consider reducing frequency to allow better recovery rather than adding more volume

[T043] IF cardio plateau coincides with poor hydration habits → **THEN** address hydration before further program changes

[T044] IF cardio plateau is reported but heart rate data shows a clear downward trend → **THEN** this is not a true plateau; reassure the person that fitness is still improving

[T045] IF cardio plateau is reported alongside a goal mismatch with general fitness training but elite-level expectations → **THEN** reset expectations to typical general fitness improvement rates

[T046] IF cardio plateau persists after all standard interventions → **THEN** recommend a full reassessment using Module 2 and a fresh program design

[T047] IF cardio plateau is reported only for walking pace, not overall fitness → **THEN** clarify that walking pace alone is a narrow metric; check broader fitness indicators

[T048] IF cardio plateau coincides with a recent switch to a new activity type → **THEN** allow 2-3 weeks of adaptation before judging the new activity's plateau status

[T049] IF cardio plateau is reported by someone who recently started strength training too → **THEN** note that combined new stimuli may extend the adaptation timeline temporarily

[T050] IF cardio plateau diagnosis is unclear after the full decision tree → **THEN** default to a deload week followed by structured interval reintroduction

[T158] IF cardio plateau is reported alongside a recently changed climate such as heat or altitude → **THEN** account for environmental adaptation time before judging plateau status

[T159] IF cardio plateau is reported but the person has switched activities frequently without sticking to one for 4+ weeks → **THEN** recommend committing to one primary activity for a focused block before judging plateau status

[T160] IF cardio plateau is reported alongside poor footwear or equipment condition → **THEN** rule out equipment issues before assuming a physiological cause

22.3 Strength Plateau Troubleshooting

[T051] IF strength has not improved in 4-6+ weeks → **THEN** begin strength plateau decision tree

[T052] IF strength plateau coincides with no progressive overload applied → **THEN** apply a single progressive overload variable from Module 4.9

[T053] IF strength plateau coincides with identical exercises for 8+ weeks → **THEN** introduce exercise variation while maintaining the same movement pattern

[T054] IF strength plateau coincides with declining sleep or nutrition → **THEN** address recovery factors before further program changes

[T055] IF strength plateau coincides with no deload taken in 8+ weeks → **THEN** insert a deload week immediately

[T056] IF strength plateau is specific to one movement pattern only → **THEN** review form and consider a regression check for that specific pattern

[T057] IF strength plateau persists despite overload, variation, and good recovery → **THEN** treat as a genuine plateau requiring a full program redesign

[T058] IF strength plateau is reported by a complete beginner within the first 3 weeks → **THEN** this is too early to be a true plateau; continue current program

[T059] IF strength performance has declined rather than plateaued → **THEN** investigate recovery, illness, or overtraining before assuming a normal plateau

[T060] IF strength plateau coincides with consistent form breakdown before reaching target reps → **THEN** the working load may be too high; reduce load slightly and rebuild

[T061] IF strength plateau is reported but reps in reserve are consistently high → **THEN** the working load is likely too light; increase load

[T062] IF strength plateau is reported alongside high accumulated training stress from other sources such as cardio or life → **THEN** review total weekly training and life stress load, not strength training in isolation

[T063] IF strength plateau resolves after a deload and renewed overload → **THEN** confirm accumulated fatigue was the root cause

[T064] IF strength plateau resolves after introducing exercise variation → **THEN** confirm staleness or adaptation was the root cause

[T065] IF person wants to know how long to try a fix before reassessing → **THEN** recommend a minimum 3-4 week trial before judging effectiveness

[T066] IF strength plateau coincides with inadequate protein intake → **THEN** recommend reviewing protein distribution across daily meals

[T067] IF strength plateau is reported alongside very low training frequency of once per week for that pattern → **THEN** recommend increasing frequency for that pattern to at least twice per week

[T068] IF strength plateau is reported but the person has not changed their assessment in months → **THEN** recommend a fresh Module 2 reassessment

[T069] IF strength plateau is reported with a goal mismatch between general fitness expectations and specialized strength goals → **THEN** reset expectations to typical general fitness strength progression rates

[T070] IF strength plateau persists after all standard interventions → **THEN** recommend a full reassessment and fresh program design

[T071] IF strength plateau is reported only in one specific exercise, not the whole pattern → **THEN** rotate to a different exercise variation targeting the same pattern

[T072] IF strength plateau coincides with a recent increase in cardio volume → **THEN** review whether cardio volume is now interfering with strength recovery

[T073] IF strength plateau is reported by someone training to failure every set → **THEN** recommend stopping 1-2 reps short of failure to allow better recovery between sessions

[T074] IF strength plateau diagnosis is unclear after the full decision tree → **THEN** default to a deload week followed by a fresh progressive overload block

[T075] IF strength plateau is reported alongside excellent recovery and good overload application → **THEN** this may reflect a true, expected slowing of progress rate at this training age; reassure and continue

[T161] IF strength plateau is reported alongside recent illness recovery → **THEN** expect a temporary dip and allow 1-2 weeks of gradual rebuilding before troubleshooting further

[T162] IF strength plateau is reported but grip strength specifically is limiting performance → **THEN** address grip training separately from the primary lift

[T163] IF strength plateau is reported alongside inconsistent exercise technique session to session → **THEN** standardize technique and tempo before introducing other variables

22.4 Mobility Plateau Troubleshooting

[T076] IF flexibility or mobility has not improved in 4-6+ weeks → **THEN** begin mobility plateau decision tree

[T077] IF mobility plateau coincides with inconsistent stretching practice → **THEN** address consistency before assuming a true mobility ceiling

[T078] IF mobility plateau coincides with stretching held under 20 seconds per rep → **THEN** increase hold duration to the recommended 20-30 second range

[T079] IF mobility plateau coincides with stretching only 1-2x per week → **THEN** increase frequency to 3-5x per week for the restricted area

[T080] IF mobility plateau persists despite consistent practice, duration, and frequency → **THEN** consider this may reflect an individual structural ceiling; redirect focus to functional capability within current range

[T081] IF mobility plateau is specific to one joint only → **THEN** intensify dedicated work for that specific joint per Module 8 guidance

[T082] IF mobility plateau is reported by a complete beginner within the first 2 weeks → **THEN** this is too early to be a true plateau; continue current routine

[T083] IF mobility plateau coincides with high overall life stress → **THEN** note that stress-related muscle guarding can limit flexibility gains; address stress alongside stretching

[T084] IF mobility plateau is reported in squat depth specifically → **THEN** check ankle dorsiflexion first, since it is a common limiting factor

[T085] IF mobility plateau is reported in overhead reach specifically → **THEN** check thoracic spine mobility alongside shoulder mobility

[T086] IF mobility plateau resolves after increasing stretch hold duration → **THEN** confirm insufficient dosage was the root cause

[T087] IF mobility plateau resolves after increasing frequency to daily practice → **THEN** confirm insufficient frequency was the root cause

[T088] IF person wants to know how long to try a fix before reassessing → **THEN** recommend a minimum 3-4 week trial before judging effectiveness

[T089] IF mobility plateau coincides with no PNF or assisted stretching ever attempted → **THEN** consider introducing PNF stretching for an additional stimulus

[T090] IF mobility plateau is reported alongside pain during stretching, not just tension → **THEN** stop and recommend professional evaluation rather than continued self-troubleshooting

[T091] IF mobility plateau is reported but functional movement quality has improved regardless → **THEN** redirect focus to functional improvement as the more meaningful outcome

[T092] IF mobility plateau diagnosis is unclear after the full decision tree → **THEN** default to increasing both frequency and duration of stretching simultaneously for 4 weeks

[T164] IF mobility plateau is reported alongside cold training environment → **THEN** account for reduced tissue pliability in cold conditions before judging plateau status

[T165] IF mobility plateau is reported but daily sitting time has increased recently → **THEN** address the increased sitting time as a likely contributing factor

22.5 Recovery Problems

[T093] IF soreness lasts beyond 72 hours after a session → **THEN** reduce volume or intensity for that muscle group in the next session targeting it

[T094] IF soreness is accompanied by sharp pain, not dull ache → **THEN** stop training that movement and recommend professional evaluation

[T095] IF soreness is reported after every single session regardless of intensity → **THEN** review whether adequate warm-up and gradual progression are being applied

[T096] IF soreness is severe enough to limit basic daily movement → **THEN** recommend rest and gentle active recovery until daily movement is unaffected

[T097] IF soreness resolves faster than expected after a new exercise stimulus → **THEN** confirm the repeated bout effect is occurring; this is a normal positive adaptation

[T098] IF recovery feels incomplete despite adequate sleep hours logged → **THEN** investigate sleep quality factors, not just duration

[T099] IF recovery feels incomplete despite a full rest day → **THEN** investigate whether the rest day included any genuine rest or was still highly active

[T100] IF recovery problems coincide with high alcohol consumption → **THEN** flag alcohol as a likely contributor to impaired sleep and recovery quality

[T101] IF recovery problems coincide with a recently added second daily activity such as a new sport → **THEN** review total combined training load across all activities, not just the structured program

[T102] IF recovery problems are reported only for the lower body, not upper body → **THEN** review lower-body training volume and impact specifically

[T103] IF recovery problems are reported only after high-intensity interval sessions → **THEN** ensure adequate spacing between high-intensity sessions

[T104] IF recovery problems persist despite all standard interventions → **THEN** recommend a full week of reduced training load and reassessment afterward

[T105] IF recovery problems are reported alongside resting heart rate trending upward → **THEN** treat elevated resting heart rate as confirming evidence of incomplete recovery

[T106] IF recovery problems are reported alongside mood decline or irritability → **THEN** consider this a broader overtraining signal requiring more significant rest

[T107] IF recovery problems resolve after a single extra rest day → **THEN** confirm minor accumulated fatigue was the cause; no further changes needed

[T166] IF recovery problems are reported alongside a new, unfamiliar training stimulus → **THEN** expect a temporary elevated recovery need during the adaptation period

[T167] IF recovery problems are reported alongside very high ambient heat during training → **THEN** account for additional heat-related recovery demand

[T168] IF recovery problems are reported but the person has not taken a single rest day in 3+ weeks → **THEN** mandate at least one full rest day immediately

22.6 Motivation Problems

[T108] IF motivation is low but the person has a strong recent history of consistency → **THEN** treat as a temporary dip; maintain minimum viable workouts and avoid program changes

[T109] IF motivation is low and has been declining steadily over weeks → **THEN** investigate burnout, monotony, or unaddressed life stress as root causes

[T110] IF motivation is low specifically for one type of training, not all training → **THEN** swap that activity for an alternative covering the same training need

[T111] IF motivation is low and tied to disappointment with visible progress → **THEN** redirect to non-scale progress markers and reset timeline expectations

[T112] IF motivation is low following a missed goal or event → **THEN** reframe the goal and rebuild momentum with a small, achievable near-term target

[T113] IF motivation is low and the person reports general life stress unrelated to fitness → **THEN** address overall stress load; consider reducing fitness demands temporarily

[T114] IF motivation is low and the person has no clear underlying reason for training → **THEN** help identify a deeper personal value connected to fitness

[T115] IF motivation is low despite enjoying the activity itself → **THEN** investigate external barriers such as time, energy, or logistics rather than motivation itself

[T116] IF motivation is low and the person has been training the same program for months → **THEN** introduce variety in exercises or format

[T117] IF motivation is low and the person lacks any social or accountability support → **THEN** recommend adding an accountability partner or community element

[T118] IF motivation resolves after introducing program variety → **THEN** confirm monotony was the root cause

[T119] IF motivation resolves after addressing an unrelated life stressor → **THEN** confirm external stress was the root cause, not the program itself

[T120] IF motivation problems persist despite all standard interventions → **THEN** recommend a planned short break with a clear, specific return date rather than indefinite pause

[T121] IF motivation is low and tied to comparison with others → **THEN** redirect focus exclusively to personal trend data

[T122] IF motivation is low and the person reports feeling embarrassed about their fitness level → **THEN** normalize the experience and recommend training in a more private setting if helpful

[T169] IF motivation is low and the person recently achieved a major goal → **THEN** help set a new meaningful goal to replace the completed one

[T170] IF motivation is low and tied to feeling the program no longer fits their life stage → **THEN** reassess goals and program fit using Module 18

[T171] IF motivation is high but inconsistent with actual follow-through → **THEN** investigate practical barriers separately from motivation itself

22.7 Schedule Adherence Problems

[T123] IF workouts are missed repeatedly at the same scheduled time → **THEN** the scheduled time likely does not fit; identify a more realistic time slot

[T124] IF workouts are missed due to unpredictable daily schedule changes → **THEN** switch to a flexible weekly target instead of fixed daily times

[T125] IF workouts are missed due to low energy at the scheduled time → **THEN** test an alternate time of day when energy is typically higher

[T126] IF workouts are missed due to competing social or family commitments → **THEN** build flexibility into the schedule and treat short sessions as acceptable substitutes

[T127] IF workouts are missed due to the session feeling too long or demanding → **THEN** reduce session length temporarily to rebuild adherence before increasing again

[T128] IF workouts are missed due to lack of a clear plan for each session → **THEN** provide a pre-set, simple session plan to remove decision fatigue

[T129] IF workouts are missed two sessions in a row → **THEN** treat this as a high-risk point and prioritize the very next opportunity to train

[T130] IF workouts are missed due to travel → **THEN** switch temporarily to the Traveler lifestyle guidance

[T131] IF workouts are missed due to illness → **THEN** pause structured training and resume gradually once recovered

[T132] IF workouts are missed due to genuinely having zero available time on certain days → **THEN** offer a true minimum viable workout option of one set

[T133] IF adherence improves after switching to a flexible weekly target → **THEN** confirm fixed scheduling was the barrier

[T134] IF adherence improves after reducing session length → **THEN** confirm session duration was the barrier

[T135] IF adherence problems persist despite all standard interventions → **THEN** reassess whether the person's current goal and program truly match their actual life circumstances

[T136] IF adherence is high for cardio but low for strength sessions, or vice versa → **THEN** investigate specific barriers unique to the lower-adherence activity type

[T137] IF adherence is inconsistent but overall weekly volume is still being met through other activity → **THEN** recognize this as acceptable flexible adherence rather than a true problem

[T172] IF schedule adherence is good on weekdays but poor on weekends → **THEN** build a separate, more flexible weekend session plan

[T173] IF schedule adherence drops during public holidays or festival periods → **THEN** plan a reduced-expectation holiday protocol in advance

[T174] IF schedule adherence is poor specifically due to commute time → **THEN** recommend combining commute walking or cycling with structured training

22.8 Lifestyle Obstacles

[T138] IF person reports a generally busy schedule as the main obstacle → **THEN** default to short, flexible, minimum-viable session options

[T139] IF person reports travel as the main obstacle → **THEN** assign the bodyweight hotel-room circuit and prioritize walking on travel days

[T140] IF person reports upcoming exams as the main obstacle → **THEN** reduce session count and frame remaining sessions as stress relief

[T141] IF person reports shift work as the main obstacle → **THEN** anchor sessions to personal wake/sleep schedule, not clock time

[T142] IF person reports family obligations as the main obstacle → **THEN** recommend combining activity with family time or using short windows during naps or quiet periods

[T143] IF person reports multiple overlapping obstacles simultaneously → **THEN** identify the single most dominant constraint and address that first

[T144] IF person's busy schedule is seasonal such as a year-end work crunch → **THEN** plan a temporary reduced-volume phase with a clear return-to-normal date

[T145] IF person's travel obstacle is long-term or permanent due to a frequent travel role → **THEN** build a permanent flexible, travel-friendly program rather than treating it as temporary

[T146] IF person's exam-period obstacle recurs every semester → **THEN** create a standing reduced-intensity exam-period protocol to reuse each time

[T147] IF person's shift work pattern rotates unpredictably → **THEN** use a flexible weekly target with a ready menu of session lengths

[T148] IF person's family obligations are temporary such as a newborn → **THEN** plan for a temporary significant reduction in volume with gradual return as circumstances change

[T149] IF person reports an obstacle not covered by existing categories → **THEN** apply the general principle of reducing complexity and volume while preserving minimal consistency

[T150] IF lifestyle obstacle resolves or eases after a defined period → **THEN** guide the person back toward their previous program gradually rather than all at once

[T151] IF person reports feeling guilty about lifestyle-driven reduced training → **THEN** reassure that adapting to life circumstances is a sign of sustainable practice, not failure

[T152] IF person asks how to prevent a lifestyle obstacle from becoming a permanent dropout → **THEN** emphasize maintaining the smallest possible consistent habit throughout the obstacle period

[T175] IF person reports caregiving responsibilities as the main obstacle → **THEN** recommend brief sessions during predictable caregiving windows

[T176] IF person reports working two jobs as the main obstacle → **THEN** prioritize very short, high-efficiency sessions during any available gap

[T177] IF person reports a recent major relocation as the main obstacle → **THEN** allow a temporary reduced-structure period while routines re-establish

[T178] IF person reports unreliable internet or app access affecting tracking → **THEN** default to simple paper or memory-based tracking instead

[T179] IF person reports a noise-sensitive living situation → **THEN** substitute low-impact, quiet exercise variations

[T180] IF person reports lack of privacy as the main obstacle → **THEN** recommend standing, low-visibility exercises or off-peak timing

[T181] IF person reports unsafe outdoor conditions in their area → **THEN** default to indoor bodyweight and home equipment alternatives

[T182] IF person reports financial constraints limiting equipment or gym access → **THEN** default entirely to bodyweight and outdoor training options

[T183] IF person reports a language or literacy barrier with written program materials → **THEN** simplify instructions to short, plain-language cues with minimal text

[T184] IF person reports a disability affecting standard exercise selection → **THEN** adapt exercise selection using available regressions and recommend professional guidance for significant adaptations

22.9 General Troubleshooting Meta-Rules

[T185] IF person reports multiple simultaneous troubleshooting issues across different categories → **THEN** address the issue with the most severe safety implication first

[T186] IF a troubleshooting fix shows partial but incomplete improvement → **THEN** maintain the fix and allow additional time before introducing a second change

[T187] IF a troubleshooting fix shows no improvement after the recommended trial period → **THEN** move to the next most likely cause in that category's decision tree

[T188] IF person wants to troubleshoot an issue not explicitly covered in this module → **THEN** apply the closest matching decision tree and use conservative, safety-first defaults

[T189] IF person requests an immediate fix without time for a proper trial period → **THEN** offer the lowest-risk, fastest-acting recommendation first while noting full resolution may take longer

[T190] IF troubleshooting reveals a likely medical concern rather than a training or lifestyle issue → **THEN** stop self-directed troubleshooting and route to professional evaluation

[T191] IF person successfully resolves an issue using this module → **THEN** document which fix worked for faster troubleshooting if the issue recurs

[T192] IF an issue recurs after previously being resolved → **THEN** re-apply the previously effective fix before exploring new causes

[T193] IF person is uncertain which troubleshooting category their issue belongs to → **THEN** ask which of the six core areas of sleep, stress, hydration, nutrition, recovery, or training load seems most affected

[T194] IF person reports an issue affecting multiple training qualities at once such as energy, strength, and motivation together → **THEN** prioritize the recovery and lifestyle foundations, since they typically affect all qualities simultaneously

[T195] IF person reports an issue that resolved on its own without intervention → **THEN** note this as likely a minor, self-resolving fluctuation and continue monitoring

[T196] IF person reports a troubleshooting issue lasting under 3 days → **THEN** treat as normal short-term variation rather than a pattern requiring intervention

[T197] IF person reports a troubleshooting issue lasting over 3 weeks → **THEN** treat as a confirmed pattern requiring structured troubleshooting

[T198] IF person wants to prevent a known recurring issue proactively → **THEN** build the previously effective fix into the standing weekly routine

[T199] IF person's troubleshooting issue coincides with a major positive life event such as a new job or relationship → **THEN** acknowledge the disruption as normal and allow temporary flexibility

[T200] IF person's troubleshooting issue coincides with a major negative life event → **THEN** prioritize emotional wellbeing and minimal-viable consistency over program advancement

[T201] IF person asks how this troubleshooting module relates to the FAQ database → **THEN** clarify this module provides structured decision trees while the FAQ database provides direct question-answer pairs

[T202] IF person's issue spans both physical and motivational causes simultaneously → **THEN** address the physical cause first, since resolving it often restores motivation naturally

[T203] IF person reports a troubleshooting issue that conflicts with their stated program preferences → **THEN** prioritize safety and effectiveness over stated preferences when they conflict

[T204] IF person's troubleshooting need is urgent and safety-related → **THEN** escalate to relevant Module 14 or Module 15 safety guidance immediately

[T205] IF person's troubleshooting issue is purely informational, not personally experienced → **THEN** answer using the relevant decision tree without assuming personal application

[T206] IF person wants a single fallback action when no clear cause can be identified → **THEN** default to ensuring adequate sleep, hydration, and one rest day before any other change

[T207] IF person's troubleshooting issue improves after addressing an apparently unrelated factor → **THEN** document the connection for future reference, since recovery factors often interact

[T208] IF person reports troubleshooting fatigue from too many simultaneous adjustments → **THEN** simplify to one change at a time going forward

[T209] IF person asks how often to revisit this troubleshooting module → **THEN** recommend treating it as a standing reference whenever a new issue arises

[T210] IF person's troubleshooting issue has no resolution after exhausting all relevant decision trees → **THEN** recommend a full program reset using Module 2 reassessment and fresh Module 19 program design

MODULE 23 | EXERCISE MISTAKE & CORRECTION DATABASE

108 exercise correction entries across 8 movement patterns. Each entry includes purpose, common mistakes, likely cause, coaching cues, correction strategy, regression, progression, and red flags.

23.1 Push Pattern

EXERCISE: Push-Up [Push]

Purpose: Build chest, shoulder, and triceps strength using bodyweight

Common Mistakes: Hips sagging or piking; flared elbows; partial range of motion

Likely Cause: Weak core bracing; insufficient shoulder/chest strength for full range; rushing reps

Coaching Cues: Body in a straight line head to heels; elbows at roughly 45 degrees; lower chest to within a fist of the floor

Correction Strategy: Cue a plank position before starting; slow the tempo; reduce reps to maintain form

Regression: Incline Push-Up or Wall Push-Up

Progression: Decline Push-Up or Weighted Push-Up

Red Flags: Sharp shoulder pain; numbness in the hands; wrist pain that worsens with reps

EXERCISE: Incline Push-Up [Push]

Purpose: Beginner-friendly push pattern using elevated hands to reduce load

Common Mistakes: Hands too low (too close to standard push-up difficulty); hips sagging

Likely Cause: Selecting too low an incline for current strength level

Coaching Cues: Choose a surface height where the last 2 reps of the set feel challenging but completable

Correction Strategy: Raise the incline height until form holds for the full set

Regression: Wall Push-Up

Progression: Standard Push-Up

Red Flags: Shoulder pain at the top of the movement; wrist pain

EXERCISE: Bench Press [Push]

Purpose: Build maximal horizontal pushing strength using a barbell or dumbbells

Common Mistakes: Bar path drifting forward; bouncing the bar off the chest; flared elbows

Likely Cause: Insufficient shoulder blade retraction; poor bar path control; ego loading

Coaching Cues: Retract shoulder blades and keep them pinned; lower the bar to mid-chest under control

Correction Strategy: Reduce load and groove the bar path with an empty bar or light dumbbells first

Regression: Floor Press or Push-Up

Progression: Incline Bench Press or Pause Bench Press

Red Flags: Sharp shoulder pain; inability to control the bar; numbness in the arms

EXERCISE: Overhead Press [Push]

Purpose: Build vertical pressing strength through the shoulders

Common Mistakes: Excessive lower back arch; pressing the bar forward instead of straight up

Likely Cause: Tight hip flexors/lats limiting overhead range; core not braced

Coaching Cues: Squeeze glutes to avoid arching; press in a straight line just past the face

Correction Strategy: Reduce load and rebuild range of motion with a landmine press variation

Regression: Seated Dumbbell Press or Landmine Press

Progression: Standing Barbell Press or Push Press

Red Flags: Shoulder impingement-type pain; pinching sensation at the top of the movement

EXERCISE: Dumbbell Shoulder Press [Push]

Purpose: Build shoulder pressing strength with independent arm control

Common Mistakes: Excessive back arch; uneven arm extension between sides

Likely Cause: Weak core stabilization; strength imbalance between sides

Coaching Cues: Brace the core throughout; press both arms with equal control

Correction Strategy: Reduce load on both sides to match the weaker arm's capacity

Regression: Seated variation with back support

Progression: Standing variation or single-arm press

Red Flags: Sharp shoulder pain; loss of control of the weight

EXERCISE: Diamond Push-Up [Push]

Purpose: Emphasize triceps within the push pattern using a narrow hand position

Common Mistakes: Elbows flaring wide instead of staying close to the body

Likely Cause: Insufficient triceps strength for the narrow position

Coaching Cues: Keep elbows brushing the ribs throughout the movement

Correction Strategy: Widen hand position slightly until triceps strength improves

Regression: Standard Push-Up

Progression: Decline Diamond Push-Up

Red Flags: Wrist pain; elbow pain

EXERCISE: Dips [Push]

Purpose: Build triceps and chest strength through a deep pressing range

Common Mistakes: Excessive forward lean shifting load to shoulders; too-deep range causing shoulder strain

Likely Cause: Limited shoulder mobility; insufficient triceps/chest strength

Coaching Cues: Keep torso relatively upright; lower only to a comfortable, pain-free depth

Correction Strategy: Reduce range of motion or use an assisted dip machine/band

Regression: Bench Dips with bent knees

Progression: Weighted Dips

Red Flags: Sharp anterior shoulder pain; clicking with pain

EXERCISE: Landmine Press [Push]

Purpose: Build pressing strength through a shoulder-friendly angled path

Common Mistakes: Pressing straight up instead of along the natural angled path

Likely Cause: Misunderstanding the equipment's natural arc

Coaching Cues: Press the bar away and slightly upward along its natural pivoting arc

Correction Strategy: Use lighter load and shadow the movement without weight first

Regression: Half-kneeling Landmine Press with lighter load

Progression: Standing Landmine Press, heavier load

Red Flags: Shoulder pain; loss of balance during the press

EXERCISE: Pike Push-Up [Push]

Purpose: Build shoulder pressing strength as a progression toward handstand work

Common Mistakes: Insufficient hip hinge to create the pike angle; head not reaching the floor

Likely Cause: Limited shoulder and hamstring flexibility

Coaching Cues: Walk feet closer to hands to increase the pike angle; lower head toward the floor between the hands

Correction Strategy: Reduce the pike angle by walking feet further from hands

Regression: Incline Pike Push-Up

Progression: Elevated Pike Push-Up (feet raised)

Red Flags: Sharp wrist or shoulder pain; dizziness from the inverted position

EXERCISE: Resistance Band Chest Press [Push]

Purpose: Build pressing strength using portable resistance

Common Mistakes: Band slack at the start of the rep reducing tension

Likely Cause: Incorrect anchor distance or band setup

Coaching Cues: Set anchor distance so tension is present throughout the entire range

Correction Strategy: Adjust anchor point or band tension to remove slack

Regression: Lighter resistance band

Progression: Heavier resistance band or slower tempo

Red Flags: Shoulder pain; band snapping or slipping risk if worn

EXERCISE: Machine Chest Press [Push]

Purpose: Build pressing strength with fixed, guided resistance

Common Mistakes: Seat height misaligned, causing handles to be too high or low

Likely Cause: Not adjusting the machine seat to individual proportions

Coaching Cues: Adjust seat so handles align with mid-chest height

Correction Strategy: Lower the weight and re-set the seat position

Regression: Lighter machine weight

Progression: Heavier machine weight or single-arm variation

Red Flags: Shoulder pain during the press

EXERCISE: Arnold Press [Push]

Purpose: Build shoulder strength through a rotational pressing pattern

Common Mistakes: Rushing the rotation; using momentum instead of control

Likely Cause: Insufficient shoulder strength for the rotational demand

Coaching Cues: Rotate the wrists slowly and deliberately as the weight rises

Correction Strategy: Reduce load and slow the tempo significantly

Regression: Standard Dumbbell Shoulder Press

Progression: Standing Arnold Press, heavier load

Red Flags: Sharp shoulder pain during rotation

EXERCISE: Plyo Push-Up [Push]

Purpose: Build explosive pushing power

Common Mistakes: Landing with locked elbows; insufficient core bracing on landing

Likely Cause: Attempting the exercise before mastering standard push-up strength

Coaching Cues: Land with soft, slightly bent elbows; brace the core throughout

Correction Strategy: Master 15+ standard push-ups before attempting plyometric variation

Regression: Standard Push-Up performed explosively without leaving the ground

Progression: Clapping Push-Up

Red Flags: Wrist pain on landing; loss of control on landing

EXERCISE: Cable Chest Fly [Push]

Purpose: Isolate the chest through a horizontal arcing motion

Common Mistakes: Using excessive weight that shifts load to the shoulders; locking elbows

Likely Cause: Ego loading beyond what the chest muscles can control

Coaching Cues: Maintain a slight, fixed elbow bend throughout; control the arc with the chest

Correction Strategy: Reduce weight significantly and focus on the mind-muscle connection

Regression: Resistance Band Chest Fly

Progression: Heavier cable weight, slower tempo

Red Flags: Sharp anterior shoulder pain; chest strain sensation

23.2 Pull Pattern

EXERCISE: Row (Bent-Over) [Pull]

Purpose: Build mid-back and lat strength through a horizontal pulling motion

Common Mistakes: Rounding the lower back; using momentum to swing the weight up

Likely Cause: Weak posterior chain; insufficient hip-hinge strength to hold the bent-over position

Coaching Cues: Maintain a flat back throughout; pull with the elbow, not the hand

Correction Strategy: Reduce load; perform a supported row (chest on a bench) until back strength improves

Regression: Supported Chest-Supported Row

Progression: Single-Arm Bent-Over Row, heavier load

Red Flags: Sharp lower back pain; loss of bent-over position control

EXERCISE: Standing Row (Band) [Pull]

Purpose: Build pulling strength using portable resistance

Common Mistakes: Shrugging shoulders up toward the ears instead of pulling with the back

Likely Cause: Overreliance on the trapezius instead of the lats/rhomboids

Coaching Cues: Keep shoulders down and pull the elbow straight back

Correction Strategy: Reduce band tension and re-cue shoulder position

Regression: Lighter band

Progression: Heavier band or single-arm variation

Red Flags: Neck or upper trap pain

EXERCISE: Pull-Up [Pull]

Purpose: Build maximal vertical pulling strength using bodyweight

Common Mistakes: Using momentum/kiping; partial range of motion (not reaching full extension)

Likely Cause: Insufficient relative pulling strength for full bodyweight

Coaching Cues: Pull with controlled tempo; lower to a full dead hang each rep

Correction Strategy: Use band-assisted pull-ups or a lighter assistance setting

Regression: Assisted Pull-Up (band or machine)

Progression: Weighted Pull-Up

Red Flags: Sharp shoulder pain; elbow pain at full extension

EXERCISE: Lat Pulldown [Pull]

Purpose: Build lat strength through a vertical pulling motion with adjustable resistance

Common Mistakes: Leaning back excessively; pulling behind the neck

Likely Cause: Using momentum instead of controlled lat engagement

Coaching Cues: Pull the bar to upper chest level with a slight, controlled lean back only

Correction Strategy: Reduce weight and focus on full control through the range

Regression: Lighter machine weight or assisted pulldown

Progression: Heavier weight or single-arm pulldown

Red Flags: Neck or shoulder pain; sharp pain pulling behind the neck

EXERCISE: Single-Arm Row [Pull]

Purpose: Build unilateral back strength and address side-to-side imbalances

Common Mistakes: Rotating the torso to assist the pull instead of isolating the back

Likely Cause: Using body momentum to compensate for a weaker pulling side

Coaching Cues: Keep the torso stable and square; pull with the back, not the torso rotation

Correction Strategy: Reduce load on the weaker side specifically

Regression: Supported Single-Arm Row (using a bench)

Progression: Standing Single-Arm Row, heavier load

Red Flags: Sharp lower back pain from rotation

EXERCISE: Inverted Row [Pull]

Purpose: Build pulling strength using a bar or rings at an adjustable angle

Common Mistakes: Hips sagging; not achieving full chest-to-bar range

Likely Cause: Body angle too horizontal for current strength level

Coaching Cues: Maintain a straight body line; pull chest to the bar

Correction Strategy: Raise the bar height to create a steeper, easier angle

Regression: Steep-Angle Inverted Row

Progression: Horizontal Inverted Row or feet-elevated variation

Red Flags: Shoulder pain at full extension

EXERCISE: Face Pull [Pull]

Purpose: Strengthen rear shoulder and upper back muscles for posture and shoulder health

Common Mistakes: Pulling with arms only, not engaging the rear shoulders/upper back

Likely Cause: Insufficient rear-shoulder activation, overreliance on biceps

Coaching Cues: Pull the rope toward the face leading with the elbows high

Correction Strategy: Reduce resistance and focus on the elbow-high pulling cue

Regression: Lighter band/cable resistance

Progression: Heavier resistance or slower tempo

Red Flags: Sharp shoulder pain at end range

EXERCISE: Band Pull-Apart [Pull]

Purpose: Strengthen the upper back and rear shoulders to support posture

Common Mistakes: Bending the elbows excessively instead of a straight-arm pull

Likely Cause: Substituting elbow flexion for the intended straight-arm horizontal pull

Coaching Cues: Keep arms straight and pull the band apart using the upper back

Correction Strategy: Reduce band tension and re-cue straight-arm execution

Regression: Lighter resistance band

Progression: Heavier resistance band

Red Flags: Shoulder pinching pain

EXERCISE: Renegade Row [Pull]

Purpose: Combine a plank core hold with a unilateral rowing motion

Common Mistakes: Excessive hip rotation during the row; sagging hips

Likely Cause: Insufficient core stability to resist rotation while rowing

Coaching Cues: Brace the core hard and minimize hip movement throughout the row

Correction Strategy: Widen the foot stance for more stability; reduce load

Regression: Kneeling Renegade Row

Progression: Standard stance, heavier dumbbells

Red Flags: Sharp lower back pain; loss of plank position

EXERCISE: Seated Cable Row [Pull]

Purpose: Build mid-back pulling strength with controlled, seated resistance

Common Mistakes: Excessive torso rocking to generate momentum

Likely Cause: Using body momentum instead of back muscles to move the weight

Coaching Cues: Keep the torso still and pull using the back muscles only

Correction Strategy: Reduce weight to remove the need for momentum

Regression: Lighter cable weight

Progression: Heavier weight or single-arm seated row

Red Flags: Lower back pain from rocking

EXERCISE: Towel Row (Door Anchor) [Pull]

Purpose: Build pulling strength with zero equipment using a towel and sturdy door

Common Mistakes: Door not adequately secured; unstable anchor point

Likely Cause: Improvised setup lacking a secure attachment point

Coaching Cues: Confirm the door is fully closed and the anchor point is secure before pulling

Correction Strategy: Test the setup with very light body lean before committing full bodyweight

Regression: Reduce the lean-back angle

Progression: Increase the lean-back angle for more resistance

Red Flags: Door slipping or releasing under load; falling backward

EXERCISE: Superman [Pull]

Purpose: Strengthen the lower back and posterior chain through spinal extension

Common Mistakes: Excessive neck hyperextension; jerky, uncontrolled lifting

Likely Cause: Looking up instead of keeping a neutral neck position

Coaching Cues: Keep gaze toward the floor; lift in a slow, controlled manner

Correction Strategy: Reduce the range of motion lifted

Regression: Bird Dog

Progression: Superman with arm/leg holds extended longer

Red Flags: Sharp lower back pain

EXERCISE: Reverse Fly [Pull]

Purpose: Isolate the rear shoulders and upper back

Common Mistakes: Using momentum to swing the weights instead of controlled lifting

Likely Cause: Excessive load relative to rear-shoulder strength

Coaching Cues: Lift with a slight bend in the elbows using controlled tempo

Correction Strategy: Reduce load significantly and slow the tempo

Regression: Resistance Band Reverse Fly

Progression: Heavier dumbbells, slower tempo

Red Flags: Sharp shoulder pain

23.3 Squat Pattern

EXERCISE: Squat [Squat]

Purpose: Build foundational lower-body strength using a bodyweight squat pattern

Common Mistakes: Knees caving inward; heels rising off the floor; rounding the lower back

Likely Cause: Weak glute/hip external rotation strength; limited ankle dorsiflexion; weak core bracing

Coaching Cues: Knees track over the toes; weight through the whole foot; chest stays tall

Correction Strategy: Reduce depth to a pain-free range; add ankle and hip mobility work

Regression: Chair Squat (sit-to-stand with support)

Progression: Goblet Squat or Jump Squat

Red Flags: Sharp knee pain; sharp lower back pain; loss of balance

EXERCISE: Goblet Squat [Squat]

Purpose: Add light load to the squat pattern while reinforcing upright posture

Common Mistakes: Holding the weight too low, encouraging a forward lean

Likely Cause: Insufficient core/upper-back strength to keep the weight high and close

Coaching Cues: Hold the weight at chest height, elbows pointing down inside the knees at the bottom

Correction Strategy: Reduce load and re-cue the high chest-level hold

Regression: Bodyweight Squat

Progression: Front Squat (barbell)

Red Flags: Sharp knee pain; dropping the weight unexpectedly

EXERCISE: Split Squat [Squat]

Purpose: Build unilateral leg strength and address side-to-side imbalances

Common Mistakes: Front knee traveling far past the toes; poor balance causing wobbling

Likely Cause: Stance too narrow; insufficient single-leg stability

Coaching Cues: Keep most weight through the front heel; widen stance slightly for balance

Correction Strategy: Hold onto a wall or chair for support while building stability

Regression: Supported Split Squat (wall/chair)

Progression: Bulgarian Split Squat

Red Flags: Sharp front knee pain; loss of balance and falling

EXERCISE: Chair Squat [Squat]

Purpose: Beginner entry point for the squat pattern using a chair as a depth guide

Common Mistakes: Plopping down into the chair instead of controlled lowering

Likely Cause: Insufficient eccentric control strength

Coaching Cues: Lower under control and lightly tap the chair before standing back up

Correction Strategy: Use a higher chair or add hand support on armrests

Regression: Assisted Sit-to-Stand (using arms)

Progression: Bodyweight Squat (unsupported)

Red Flags: Sharp knee or hip pain

EXERCISE: Box Squat [Squat]

Purpose: Build squat strength with a controlled depth reference point

Common Mistakes: Sitting back too aggressively, losing balance backward off the box

Likely Cause: Excessive forward lean compensation or fear of the box

Coaching Cues: Sit back under control and pause briefly on the box before standing

Correction Strategy: Use a higher box and reduce load

Regression: Chair Squat

Progression: Standard Box Squat, lower box height

Red Flags: Sharp lower back pain; loss of balance backward

EXERCISE: Front Squat [Squat]

Purpose: Build quad-dominant squat strength with an anterior load position

Common Mistakes: Elbows dropping, causing the bar to roll forward

Likely Cause: Limited wrist/shoulder mobility for the front rack position

Coaching Cues: Keep elbows high and parallel to the floor throughout

Correction Strategy: Use a cross-arm grip or reduce load while building front rack mobility

Regression: Goblet Squat

Progression: Front Squat, heavier load

Red Flags: Bar rolling off the shoulders; sharp wrist pain

EXERCISE: Back Squat [Squat]

Purpose: Build maximal lower-body strength with a posterior-loaded barbell

Common Mistakes: Excessive forward lean; knees caving

Likely Cause: Weak core bracing; limited ankle/hip mobility; bar position too high or low

Coaching Cues: Brace the core hard; keep the bar secure on the upper back; track knees over toes

Correction Strategy: Reduce load and rebuild positional control with goblet squats first

Regression: Goblet Squat

Progression: Back Squat, heavier load or pause variation

Red Flags: Sharp lower back pain; knees caving with pain; loss of bar control

EXERCISE: Sumo Squat [Squat]

Purpose: Target inner thighs and glutes using a wide-stance squat variation

Common Mistakes: Knees caving inward despite the wide stance

Likely Cause: Insufficient hip external rotation strength to match the wide stance

Coaching Cues: Actively push the knees outward in line with the toes throughout

Correction Strategy: Narrow the stance slightly and reduce load

Regression: Sumo Squat, bodyweight only

Progression: Sumo Squat with added load

Red Flags: Sharp inner knee or hip pain

EXERCISE: Wall Sit [Squat]

Purpose: Build isometric squat endurance against a wall

Common Mistakes: Knees not at a 90-degree angle; feet too close or far from the wall

Likely Cause: Incorrect foot placement relative to the wall

Coaching Cues: Position feet so knees form roughly a 90-degree angle directly above ankles

Correction Strategy: Reduce hold time and adjust foot position

Regression: Higher wall sit angle (less than 90 degrees)

Progression: Wall Sit, longer duration or single-leg variation

Red Flags: Sharp knee pain; excessive shaking beyond normal fatigue

EXERCISE: Pistol Squat Progression [Squat]

Purpose: Build advanced single-leg squat strength and balance

Common Mistakes: Losing balance and falling backward; not controlling the descent

Likely Cause: Insufficient single-leg strength, balance, or ankle mobility for the full movement

Coaching Cues: Use a support for balance initially; control the descent on the working leg

Correction Strategy: Use assisted variations with a TRX or chair support

Regression: Bulgarian Split Squat

Progression: Full Unassisted Pistol Squat

Red Flags: Sharp knee pain; loss of balance and falling

EXERCISE: Cossack Squat [Squat]

Purpose: Build lateral hip and adductor mobility combined with strength

Common Mistakes: Heel of the working leg lifting off the floor; rounding the back

Likely Cause: Limited ankle and hip mobility for the deep lateral position

Coaching Cues: Keep the working foot flat; chest up throughout the movement

Correction Strategy: Reduce depth and use a support for balance

Regression: Lateral Lunge (shallower range)

Progression: Cossack Squat, full depth with added load

Red Flags: Sharp groin or knee pain

EXERCISE: Jump Squat [Squat]

Purpose: Build explosive lower-body power

Common Mistakes: Landing with locked knees; insufficient depth on landing absorption

Likely Cause: Attempting the exercise before mastering standard squat strength and control

Coaching Cues: Land softly with bent knees, absorbing impact through the whole foot

Correction Strategy: Master 15+ bodyweight squats before attempting jump squats

Regression: Bodyweight Squat performed explosively without leaving the ground

Progression: Weighted Jump Squat

Red Flags: Sharp knee pain on landing; ankle instability

EXERCISE: Step-Up [Squat]

Purpose: Build unilateral leg strength using a step or box

Common Mistakes: Pushing off the trailing leg instead of driving through the working leg

Likely Cause: Insufficient single-leg strength causing reliance on the back leg

Coaching Cues: Drive entirely through the working leg's heel; minimize trailing leg push

Correction Strategy: Lower the step height to reduce difficulty

Regression: Low Step-Up

Progression: Higher Step-Up or Weighted Step-Up

Red Flags: Sharp knee pain at the top of the step

23.4 Lunge Pattern

EXERCISE: Lunge [Lunge]

Purpose: Build unilateral leg strength and balance through a stepping pattern

Common Mistakes: Front knee traveling far past the toes; torso leaning excessively forward

Likely Cause: Insufficient stride length; weak core bracing

Coaching Cues: Take a stride long enough that the front knee stays roughly above the ankle

Correction Strategy: Shorten the stride length and reduce depth

Regression: Static Lunge (supported by a wall/chair)

Progression: Walking Lunge or Weighted Lunge

Red Flags: Sharp front knee pain; loss of balance

EXERCISE: Static Lunge [Lunge]

Purpose: Build lunge strength and stability without a stepping transition

Common Mistakes: Excessive forward lean; poor knee tracking

Likely Cause: Insufficient core bracing or hip stability

Coaching Cues: Keep the torso upright; lower straight down rather than forward

Correction Strategy: Hold a wall or chair for balance support

Regression: Supported Static Lunge

Progression: Static Lunge, unsupported with added load

Red Flags: Sharp knee pain; loss of balance

EXERCISE: Reverse Lunge [Lunge]

Purpose: Build lunge strength with reduced knee stress via a backward step

Common Mistakes: Stepping too short, increasing front knee stress

Likely Cause: Insufficient stride length on the backward step

Coaching Cues: Step back far enough that the front shin stays close to vertical

Correction Strategy: Shorten the range of motion and use a support for balance

Regression: Supported Reverse Lunge

Progression: Reverse Lunge with added load

Red Flags: Sharp front knee pain

EXERCISE: Walking Lunge [Lunge]

Purpose: Build dynamic unilateral leg strength through continuous stepping

Common Mistakes: Losing balance between steps; insufficient stride length

Likely Cause: Rushing the transition between steps before establishing stability

Coaching Cues: Pause briefly in each lunge position before stepping to the next

Correction Strategy: Reduce the number of consecutive steps and rest between sets

Regression: Reverse Lunge (stationary)

Progression: Walking Lunge with added load

Red Flags: Sharp knee pain; repeated loss of balance

EXERCISE: Bulgarian Split Squat [Lunge]

Purpose: Build advanced unilateral leg strength using a rear-foot-elevated position

Common Mistakes: Front knee caving inward; insufficient depth

Likely Cause: Weak hip stability on the working leg; rear foot elevation too high for current strength

Coaching Cues: Track the front knee in line with the toes; lower the rear foot elevation if needed

Correction Strategy: Lower the rear foot elevation height or reduce load

Regression: Static Lunge (no elevation)

Progression: Bulgarian Split Squat, heavier load

Red Flags: Sharp front knee pain; loss of balance and falling

EXERCISE: Lateral Lunge [Lunge]

Purpose: Build frontal-plane hip and adductor strength

Common Mistakes: Rounding the back; insufficient hip hinge on the working side

Likely Cause: Limited hip mobility for the lateral movement

Coaching Cues: Hinge at the hips while stepping laterally; keep the trailing leg straight

Correction Strategy: Reduce the lateral step distance

Regression: Lateral Step (shallow range)

Progression: Lateral Lunge with added load

Red Flags: Sharp groin or knee pain

EXERCISE: Curtsy Lunge [Lunge]

Purpose: Target glutes through a cross-behind stepping pattern

Common Mistakes: Twisting the torso instead of keeping it square

Likely Cause: Insufficient hip stability for the rotational stepping pattern

Coaching Cues: Keep the torso facing forward throughout; step the leg behind and across

Correction Strategy: Reduce the depth and range of the crossing step

Regression: Reverse Lunge (standard)

Progression: Curtsy Lunge with added load

Red Flags: Sharp knee pain; loss of balance

EXERCISE: Lunge with Twist [Lunge]

Purpose: Combine lunge strength with rotational core engagement

Common Mistakes: Twisting from the lower back instead of the upper torso

Likely Cause: Insufficient core control to isolate rotation to the torso

Coaching Cues: Rotate from the ribcage and shoulders, keeping the hips relatively stable

Correction Strategy: Remove the twist and master the standard lunge first

Regression: Static Lunge (no twist)

Progression: Lunge with Twist, added resistance

Red Flags: Sharp lower back pain during rotation

EXERCISE: Deficit Reverse Lunge [Lunge]

Purpose: Increase range of motion in the lunge pattern using an elevated platform

Common Mistakes: Insufficient depth control on the way down; losing balance

Likely Cause: Attempting too high a deficit before building base lunge strength

Coaching Cues: Start with a small deficit height and control the descent fully

Correction Strategy: Reduce or remove the deficit height

Regression: Standard Reverse Lunge

Progression: Deficit Reverse Lunge, greater height or added load

Red Flags: Sharp knee pain; loss of balance

EXERCISE: Lunge Jump [Lunge]

Purpose: Build explosive unilateral power through an alternating jumping lunge

Common Mistakes: Landing with knees caving; insufficient control on landing

Likely Cause: Attempting before mastering standard walking lunge strength

Coaching Cues: Land softly with control, knees tracking over toes on each side

Correction Strategy: Master 12+ walking lunges per side before attempting jump lunges

Regression: Walking Lunge (no jump)

Progression: Weighted Lunge Jump

Red Flags: Sharp knee pain on landing; ankle instability

EXERCISE: Clock Lunge [Lunge]

Purpose: Build multi-directional lunge strength and balance

Common Mistakes: Losing balance when stepping to non-forward directions

Likely Cause: Limited practice with lateral and rotational stepping patterns

Coaching Cues: Move slowly through each direction, pausing briefly to regain balance

Correction Strategy: Reduce the number of directions practiced per set

Regression: Reverse Lunge (single direction)

Progression: Full Clock Lunge with added load

Red Flags: Sharp knee or ankle pain; repeated loss of balance

EXERCISE: Lunge to Knee Drive [Lunge]

Purpose: Combine lunge strength with balance and hip flexor activation

Common Mistakes: Rushing the knee drive without controlling the lunge portion first

Likely Cause: Insufficient single-leg balance to control both phases of the movement

Coaching Cues: Complete the lunge fully before driving the knee up with control

Correction Strategy: Separate the two phases initially: master the lunge, then add the knee drive

Regression: Reverse Lunge (no knee drive)

Progression: Lunge to Knee Drive with a balance hold at the top

Red Flags: Loss of balance and falling

23.5 Hinge Pattern

EXERCISE: Romanian Deadlift [Hinge]

Purpose: Build posterior chain strength through a hip-dominant hinge with a slight knee bend

Common Mistakes: Rounding the lower back; bending the knees too much, turning it into a squat

Likely Cause: Weak hamstring flexibility; insufficient core bracing to maintain a flat back

Coaching Cues: Push the hips back like closing a car door; keep a flat back like a tabletop

Correction Strategy: Reduce range of motion and load until hip-hinge mechanics improve

Regression: Glute Bridge

Progression: Romanian Deadlift, heavier load or single-leg variation

Red Flags: Sharp lower back pain; numbness or tingling down the legs

EXERCISE: Deadlift [Hinge]

Purpose: Build maximal posterior chain and full-body pulling strength from the floor

Common Mistakes: Rounding the back; bar drifting away from the body; hips rising faster than the chest

Likely Cause: Weak core bracing; poor bar path awareness; insufficient hip/hamstring strength

Coaching Cues: Keep the bar close to the shins/thighs throughout; chest and hips rise together

Correction Strategy: Reduce load and rebuild the pattern with Romanian deadlifts or trap-bar variation

Regression: Romanian Deadlift (reduced range)

Progression: Deadlift, heavier load

Red Flags: Sharp lower back pain; numbness or tingling; loss of bar control

EXERCISE: Hip Bridge [Hinge]

Purpose: Build foundational glute and hip extension strength

Common Mistakes: Overarching the lower back instead of extending through the hips

Likely Cause: Weak glute activation, overreliance on the lower back to create extension

Coaching Cues: Squeeze the glutes hard at the top; avoid excessive lower back arch

Correction Strategy: Reduce range of motion and focus on the glute squeeze cue

Regression: Glute Bridge with bent knees, partial range

Progression: Single-Leg Glute Bridge or Hip Thrust

Red Flags: Sharp lower back pain

EXERCISE: Hip Thrust [Hinge]

Purpose: Build advanced glute strength using a shoulder-elevated hip extension

Common Mistakes: Hyperextending the lower back at the top instead of stopping at full hip extension

Likely Cause: Overshooting the range due to seeking more visible extension

Coaching Cues: Stop the movement when hips reach full extension; avoid lower back arch

Correction Strategy: Reduce load and range of motion

Regression: Hip Bridge (floor-based)

Progression: Hip Thrust, heavier load or single-leg

Red Flags: Sharp lower back pain

EXERCISE: Single-Leg Glute Bridge [Hinge]

Purpose: Build unilateral glute strength and address side imbalances

Common Mistakes: Hips rotating or dropping on the non-working side

Likely Cause: Insufficient hip stability on the working side

Coaching Cues: Keep hips level and square throughout; squeeze the working glute

Correction Strategy: Reduce range of motion or add light support under the resting leg

Regression: Standard Glute Bridge (both legs)

Progression: Single-Leg Glute Bridge with added load

Red Flags: Sharp lower back or hip pain

EXERCISE: Good Morning [Hinge]

Purpose: Build posterior chain strength with a barbell-loaded hinge from standing

Common Mistakes: Rounding the back; bending the knees excessively

Likely Cause: Weak hamstring flexibility; insufficient core bracing under load

Coaching Cues: Maintain a flat back and slight knee bend; hinge from the hips

Correction Strategy: Reduce load significantly and use bodyweight hinges to rebuild the pattern

Regression: Bodyweight Hip Hinge

Progression: Good Morning, heavier load

Red Flags: Sharp lower back pain

EXERCISE: Kettlebell Swing [Hinge]

Purpose: Build explosive hip-hinge power and conditioning

Common Mistakes: Squatting the movement instead of hinging; using the arms to lift the weight

Likely Cause: Misunderstanding the hip-drive nature of the movement

Coaching Cues: Drive through the hips explosively; let the arms simply guide the kettlebell

Correction Strategy: Reduce weight and rebuild the hinge pattern without the kettlebell first

Regression: Bodyweight Hip Hinge

Progression: Kettlebell Swing, heavier load

Red Flags: Sharp lower back pain; losing grip on the kettlebell

EXERCISE: Single-Leg Romanian Deadlift [Hinge]

Purpose: Build unilateral posterior chain strength and balance

Common Mistakes: Hips rotating open instead of staying square

Likely Cause: Insufficient hip stability and balance on the standing leg

Coaching Cues: Keep the hips square to the floor throughout; move slowly and with control

Correction Strategy: Hold a wall or chair for balance support, reduce range of motion

Regression: Supported Single-Leg RDL

Progression: Single-Leg RDL with added load

Red Flags: Sharp lower back pain; loss of balance and falling

EXERCISE: Back Extension [Hinge]

Purpose: Strengthen the lower back and posterior chain through controlled spinal extension

Common Mistakes: Hyperextending beyond a neutral spine position at the top

Likely Cause: Overshooting the range seeking more visible movement

Coaching Cues: Rise only to a neutral, flat-back position, not beyond

Correction Strategy: Reduce range of motion and focus on control

Regression: Bird Dog

Progression: Back Extension with added load or slower tempo

Red Flags: Sharp lower back pain

EXERCISE: Trap Bar Deadlift [Hinge]

Purpose: Build deadlift strength with a more upright, joint-friendly torso angle

Common Mistakes: Squatting the movement excessively instead of hinging

Likely Cause: Misunderstanding the hybrid squat-hinge nature of the trap bar position

Coaching Cues: Maintain a moderate forward torso lean while driving through the legs and hips

Correction Strategy: Reduce load and rebuild the hinge pattern with bodyweight first

Regression: Romanian Deadlift

Progression: Trap Bar Deadlift, heavier load

Red Flags: Sharp lower back pain; numbness or tingling down the legs

EXERCISE: Cable Pull-Through [Hinge]

Purpose: Reinforce hip-hinge mechanics using cable resistance

Common Mistakes: Squatting instead of hinging; insufficient hip drive

Likely Cause: Misunderstanding the hip-dominant nature of the movement

Coaching Cues: Push the hips back to load the hamstrings, then drive forward through the hips

Correction Strategy: Reduce cable resistance and focus purely on the hinge pattern

Regression: Bodyweight Hip Hinge

Progression: Cable Pull-Through, heavier resistance

Red Flags: Sharp lower back pain

EXERCISE: Reverse Hyperextension [Hinge]

Purpose: Build posterior chain strength through a leg-driven hip extension

Common Mistakes: Using momentum to swing the legs instead of controlled hip extension

Likely Cause: Insufficient core/glute strength for controlled execution

Coaching Cues: Lift the legs with control using the glutes, avoiding swinging

Correction Strategy: Reduce range of motion and remove momentum

Regression: Bird Dog

Progression: Reverse Hyperextension with added load

Red Flags: Sharp lower back pain

23.6 Core Pattern

EXERCISE: Plank [Core]

Purpose: Build foundational core stability and endurance

Common Mistakes: Hips sagging toward the floor; hips piked too high

Likely Cause: Weak core bracing; insufficient shoulder stability to maintain the position

Coaching Cues: Maintain a straight line from head to heels; brace the core as if expecting a gentle poke

Correction Strategy: Reduce hold time and regress to a knee or wall plank

Regression: Knee Plank or Wall Plank

Progression: Long-Lever Plank or Weighted Plank

Red Flags: Sharp lower back pain; shoulder pain

EXERCISE: Side Plank [Core]

Purpose: Build lateral core stability and address side-to-side imbalances

Common Mistakes: Hips dropping toward the floor; rotating the torso forward or backward

Likely Cause: Weak lateral core (oblique) strength on the working side

Coaching Cues: Stack the hips and shoulders vertically; lift the hips as high as comfortably possible

Correction Strategy: Reduce hold time or bend the bottom knee for added support

Regression: Knee-Supported Side Plank

Progression: Side Plank with leg lift or added load

Red Flags: Sharp lower back or shoulder pain

EXERCISE: Bird Dog [Core]

Purpose: Build core stability and coordination through opposite-limb extension

Common Mistakes: Rotating the hips or rounding the back during limb extension

Likely Cause: Insufficient core bracing to resist rotation while extending limbs

Coaching Cues: Keep hips and shoulders square to the floor; extend limbs slowly with control

Correction Strategy: Reduce range of motion or perform without simultaneous arm/leg extension

Regression: Arm-Only or Leg-Only Extension

Progression: Bird Dog with added resistance band

Red Flags: Sharp lower back pain

EXERCISE: Mountain Climber [Core]

Purpose: Build dynamic core stability combined with a cardio element

Common Mistakes: Hips bouncing up and down instead of staying level

Likely Cause: Moving too fast before establishing core control

Coaching Cues: Keep hips level and stable; slow the tempo if control is lost

Correction Strategy: Slow the tempo significantly or reduce range of motion

Regression: Slow Mountain Climber

Progression: Mountain Climber, faster tempo or added incline

Red Flags: Sharp lower back or wrist pain

EXERCISE: Dead Bug [Core]

Purpose: Build core stability while maintaining a stable lower back position

Common Mistakes: Lower back arching off the floor during limb extension

Likely Cause: Insufficient core bracing to maintain contact between the back and the floor

Coaching Cues: Press the lower back into the floor throughout; extend limbs only as far as that position allows

Correction Strategy: Reduce range of motion of the extending limb

Regression: Dead Bug, bent-knee only without full extension

Progression: Dead Bug with added resistance

Red Flags: Sharp lower back pain

EXERCISE: Pallof Press [Core]

Purpose: Build anti-rotational core strength using cable or band resistance

Common Mistakes: Allowing the torso to rotate toward the resistance instead of resisting it

Likely Cause: Insufficient core strength relative to the resistance level used

Coaching Cues: Brace hard and press straight out, resisting any rotation of the torso

Correction Strategy: Reduce resistance and re-cue the anti-rotation brace

Regression: Pallof Press, lighter resistance

Progression: Pallof Press, heavier resistance or kneeling variation

Red Flags: Sharp lower back pain

EXERCISE: Hollow Body Hold [Core]

Purpose: Build advanced core control through a full-body braced position

Common Mistakes: Lower back arching off the floor; holding the breath

Likely Cause: Insufficient core strength for the demanding lever length used

Coaching Cues: Press the lower back firmly into the floor; breathe steadily throughout

Correction Strategy: Bend the knees and reduce arm extension to shorten the lever

Regression: Tucked Hollow Hold with knees bent and arms by sides

Progression: Full Hollow Body Hold, longer duration

Red Flags: Sharp lower back pain

EXERCISE: Russian Twist [Core]

Purpose: Build rotational core strength and control

Common Mistakes: Moving quickly with momentum rather than controlled rotation

Likely Cause: Insufficient core strength relative to speed/load used

Coaching Cues: Rotate slowly and with control, touching the floor lightly on each side

Correction Strategy: Slow the tempo and reduce or remove added weight

Regression: Seated Twist (feet on floor)

Progression: Russian Twist, feet elevated or added weight

Red Flags: Sharp lower back pain

EXERCISE: Ab Wheel Rollout [Core]

Purpose: Build advanced core strength through an extended-lever rolling motion

Common Mistakes: Lower back sagging during the extended position

Likely Cause: Attempting too long a rollout range before building sufficient core strength

Coaching Cues: Brace hard throughout; only roll out as far as a flat back can be maintained

Correction Strategy: Reduce the rollout distance significantly or perform from the knees with limited range

Regression: Plank or Dead Bug to build foundational strength first

Progression: Ab Wheel Rollout, full extension or standing variation

Red Flags: Sharp lower back pain

EXERCISE: Hanging Leg Raise [Core]

Purpose: Build advanced lower-core strength using a hanging position

Common Mistakes: Using momentum/swinging instead of controlled leg lifting

Likely Cause: Insufficient core strength relative to the demanding lever length

Coaching Cues: Lift the legs slowly and with control, avoiding any swinging

Correction Strategy: Bend the knees to shorten the lever, or reduce range of motion

Regression: Lying Leg Raise (floor-based)

Progression: Hanging Leg Raise, straight legs or added ankle weight

Red Flags: Sharp lower back pain; shoulder strain from grip

EXERCISE: Sit-Up [Core]

Purpose: Build foundational core flexion strength

Common Mistakes: Pulling on the neck with the hands; using momentum to swing up

Likely Cause: Insufficient core strength leading to compensatory neck pulling

Coaching Cues: Keep hands lightly at the temples or crossed on the chest; lift with the core

Correction Strategy: Reduce range of motion to a crunch instead of a full sit-up

Regression: Crunch (partial range)

Progression: Sit-Up with added resistance

Red Flags: Sharp neck or lower back pain

EXERCISE: Plank with Shoulder Tap [Core]

Purpose: Build anti-rotational core stability in a plank position

Common Mistakes: Hips rotating or swaying with each shoulder tap

Likely Cause: Insufficient anti-rotation core strength for the added arm movement

Coaching Cues: Widen the foot stance for stability; minimize hip movement during each tap

Correction Strategy: Widen stance further or slow the tempo significantly

Regression: Standard Plank (no taps)

Progression: Plank with Shoulder Tap, narrower stance or faster tempo

Red Flags: Sharp lower back pain; loss of plank position

EXERCISE: Stir the Pot [Core]

Purpose: Build advanced core stability using an unstable surface such as a stability ball

Common Mistakes: Excessive hip movement instead of controlled circular arm motion

Likely Cause: Insufficient core bracing for the unstable surface demand

Coaching Cues: Brace hard and keep hips still while moving only the arms in a controlled circle

Correction Strategy: Reduce the circle size or perform a standard plank on the ball first

Regression: Forearm Plank on Stability Ball (no movement)

Progression: Stir the Pot, larger circles or added load

Red Flags: Sharp lower back pain; loss of balance off the ball

23.7 Cardio Pattern

EXERCISE: Walking [Cardio]

Purpose: Build foundational cardiovascular fitness with minimal joint stress

Common Mistakes: Excessively slow pace providing minimal cardiovascular stimulus; poor posture

Likely Cause: Underestimating the pace needed for meaningful cardio benefit

Coaching Cues: Walk at a pace where conversation is possible but slightly effortful; keep an upright posture

Correction Strategy: Gradually increase pace or add an incline for more stimulus

Regression: Slower, shorter-duration walk

Progression: Brisk Walking or Walk-Jog Intervals

Red Flags: Chest pain; severe shortness of breath; dizziness

EXERCISE: Jogging [Cardio]

Purpose: Build cardiovascular endurance through continuous moderate-impact running

Common Mistakes: Overstriding by landing far ahead of the body; starting too fast and burning out

Likely Cause: Poor running mechanics; pacing without experience

Coaching Cues: Land with the foot roughly under the hips; start conservatively and build pace gradually

Correction Strategy: Reduce pace and insert walk breaks using run-walk intervals

Regression: Run-Walk Intervals

Progression: Continuous Jogging, longer duration or faster pace

Red Flags: Sharp shin or knee pain; chest pain; severe breathlessness

EXERCISE: Cycling [Cardio]

Purpose: Build cardiovascular fitness with minimal joint impact

Common Mistakes: Saddle height too low or high, causing knee strain

Likely Cause: Incorrect bike fit

Coaching Cues: Adjust saddle height so the knee has a slight bend at the bottom of the pedal stroke

Correction Strategy: Reassess and correct saddle height and position

Regression: Lower-intensity, shorter duration cycling

Progression: Cycling, higher resistance or longer duration

Red Flags: Sharp knee pain; numbness in hands or seat area

EXERCISE: Jump Rope [Cardio]

Purpose: Build cardiovascular fitness and coordination with a time-efficient, high-impact activity

Common Mistakes: Jumping too high; landing flat-footed instead of on the balls of the feet

Likely Cause: Excessive vertical jump height; poor landing mechanics

Coaching Cues: Keep jumps low and quick; land softly on the balls of the feet

Correction Strategy: Practice without the rope first by mimicking the motion to build rhythm

Regression: Bodyweight Marching or Low-Impact Step

Progression: Jump Rope, faster pace or double-unders

Red Flags: Sharp shin or ankle pain; knee pain

EXERCISE: Stair Climbing [Cardio]

Purpose: Build cardiovascular fitness and lower-body strength using stairs

Common Mistakes: Leaning too far forward; not using the full foot on each step

Likely Cause: Rushing the pace beyond current fitness level

Coaching Cues: Keep the torso upright; place the whole foot on each step where possible

Correction Strategy: Reduce pace or use a handrail for balance support

Regression: Slower pace stair climbing

Progression: Stair Climbing, faster pace or two steps at a time

Red Flags: Sharp knee pain; dizziness; chest pain

EXERCISE: Swimming [Cardio]

Purpose: Build cardiovascular fitness with near-zero joint impact

Common Mistakes: Holding the breath instead of rhythmic breathing; poor body position causing drag

Likely Cause: Unfamiliarity with breathing technique and body positioning

Coaching Cues: Establish a consistent breathing rhythm; keep the body horizontal and streamlined

Correction Strategy: Reduce distance and duration and focus on technique with a kickboard

Regression: Shorter duration, technique-focused swimming

Progression: Swimming, longer duration or faster pace

Red Flags: Chest pain; severe breathlessness; disorientation in water

EXERCISE: Elliptical Training [Cardio]

Purpose: Build cardiovascular fitness with low-impact, machine-guided movement

Common Mistakes: Leaning heavily on the handles, reducing leg engagement

Likely Cause: Using the handles as a crutch rather than for light balance only

Coaching Cues: Keep most of the effort in the legs; use the handles lightly for balance only

Correction Strategy: Reduce resistance and re-cue minimal hand reliance

Regression: Lower resistance, shorter duration

Progression: Elliptical, higher resistance or longer duration

Red Flags: Sharp knee pain; dizziness

EXERCISE: Rowing Machine [Cardio]

Purpose: Build full-body cardiovascular and strength endurance

Common Mistakes: Pulling with the arms before driving with the legs

Likely Cause: Incorrect sequencing of the rowing stroke

Coaching Cues: Drive with the legs first, then lean back, then pull with the arms last

Correction Strategy: Reduce resistance and rehearse the leg-back-arms sequence slowly

Regression: Lower resistance, shorter duration

Progression: Rowing Machine, higher resistance or longer duration

Red Flags: Sharp lower back pain

EXERCISE: High Knees [Cardio]

Purpose: Build cardiovascular intensity and hip flexor activation

Common Mistakes: Leaning back excessively; insufficient knee height

Likely Cause: Compensating for fatigue by reducing range and leaning back

Coaching Cues: Drive knees up toward hip height; keep the torso upright

Correction Strategy: Reduce pace and focus on form over speed

Regression: Marching in Place

Progression: High Knees, faster pace or longer duration

Red Flags: Sharp hip flexor pain; ankle instability

EXERCISE: Burpees [Cardio]

Purpose: Build full-body cardiovascular and strength conditioning

Common Mistakes: Rounding the back during the squat-to-plank transition; sloppy push-up form

Likely Cause: Rushing the movement beyond current fitness and strength level

Coaching Cues: Move through each phase with control; maintain a flat back during transitions

Correction Strategy: Remove the push-up and/or jump components to reduce difficulty

Regression: Step-Back Burpee with no jump or push-up

Progression: Full Burpee with push-up and jump

Red Flags: Sharp lower back or wrist pain; chest pain

23.8 Mobility Pattern

EXERCISE: Cat-Cow [Mobility]

Purpose: Improve spinal mobility through flexion and extension

Common Mistakes: Moving too quickly without full range of motion

Likely Cause: Rushing through repetitions rather than moving deliberately

Coaching Cues: Move slowly, fully arching and rounding the spine through the complete range

Correction Strategy: Slow the tempo and reduce range slightly if discomfort occurs

Regression: Smaller range Cat-Cow

Progression: Cat-Cow with longer holds at each end range

Red Flags: Sharp lower back pain

EXERCISE: 90/90 Hip Switch [Mobility]

Purpose: Improve hip internal and external rotation mobility

Common Mistakes: Using the hands to push through the transition instead of hip control

Likely Cause: Insufficient hip mobility or strength to control the transition unassisted

Coaching Cues: Control the transition using the hips and core, minimizing hand assistance

Correction Strategy: Allow hand assistance initially and reduce it gradually over time

Regression: Hand-Assisted 90/90 Switch

Progression: Unassisted 90/90 Switch, added hold time

Red Flags: Sharp hip pain

EXERCISE: Doorway Chest Stretch [Mobility]

Purpose: Improve chest and anterior shoulder flexibility

Common Mistakes: Overarching the lower back to increase the stretch sensation

Likely Cause: Seeking a deeper stretch through compensation rather than true chest flexibility

Coaching Cues: Keep the core braced and avoid arching; let the chest alone provide the stretch

Correction Strategy: Reduce the stretch intensity by stepping less far through the doorway

Regression: Gentler doorway stretch angle

Progression: Deeper stretch angle, longer hold time

Red Flags: Sharp shoulder or chest pain

EXERCISE: Thoracic Rotation (Open Book) [Mobility]

Purpose: Improve thoracic spine rotational mobility

Common Mistakes: Rotating from the lower back or hips instead of the upper spine

Likely Cause: Insufficient core stability to isolate rotation to the thoracic spine

Coaching Cues: Keep the hips stacked and stable; rotate only the upper back and shoulders

Correction Strategy: Reduce rotation range and focus on isolating the thoracic spine

Regression: Smaller range Open Book

Progression: Open Book with added hold time or resistance

Red Flags: Sharp lower back pain

EXERCISE: Deep Squat Hold [Mobility]

Purpose: Improve ankle, hip, and overall lower-body mobility

Common Mistakes: Heels lifting off the floor; rounding the lower back

Likely Cause: Limited ankle dorsiflexion or hip mobility for the full depth

Coaching Cues: Keep heels flat; use a support to hold if needed rather than rounding the back

Correction Strategy: Hold onto a support such as a pole or doorframe to maintain better positioning

Regression: Supported Deep Squat Hold

Progression: Unsupported Deep Squat Hold, longer duration

Red Flags: Sharp knee or lower back pain

EXERCISE: Knee-to-Wall Ankle Drill [Mobility]

Purpose: Improve ankle dorsiflexion specifically

Common Mistakes: Heel lifting off the floor to compensate for limited range

Likely Cause: Insufficient ankle dorsiflexion range to keep the heel down

Coaching Cues: Keep the heel flat throughout; move the foot closer to the wall only as range allows

Correction Strategy: Move the starting foot position further from the wall

Regression: Knee-to-Wall Drill, greater starting distance

Progression: Knee-to-Wall Drill, minimal starting distance

Red Flags: Sharp ankle pain

EXERCISE: World's Greatest Stretch [Mobility]

Purpose: Improve full-body mobility across hips, spine, and shoulders in one sequence

Common Mistakes: Rushing through the sequence without full range at each position

Likely Cause: Treating it as a single fast movement rather than a held, multi-position stretch

Coaching Cues: Pause briefly at each position such as the lunge and rotation before transitioning

Correction Strategy: Reduce the number of positions held per side initially

Regression: Simplified sequence with lunge and rotation only

Progression: Full sequence with longer holds each position

Red Flags: Sharp hip or lower back pain

EXERCISE: Hip Flexor Stretch (Kneeling) [Mobility]

Purpose: Improve hip flexor flexibility, particularly relevant for prolonged sitting

Common Mistakes: Overarching the lower back instead of isolating the hip flexor stretch

Likely Cause: Compensating with lumbar extension rather than true hip extension

Coaching Cues: Tuck the pelvis slightly and squeeze the glute on the back leg to isolate the stretch

Correction Strategy: Reduce the stretch depth and focus on the pelvic tuck cue

Regression: Standing Hip Flexor Stretch

Progression: Kneeling Hip Flexor Stretch, added overhead reach

Red Flags: Sharp hip or lower back pain

EXERCISE: Standing Toe Touch [Mobility]

Purpose: Improve hamstring and lower back flexibility

Common Mistakes: Bouncing into the stretch; rounding aggressively from the lower back with momentum

Likely Cause: Using momentum, known as ballistic stretching, instead of a controlled static hold

Coaching Cues: Lower slowly and hold at a point of mild tension without bouncing

Correction Strategy: Bend the knees slightly to reduce hamstring demand

Regression: Standing Toe Touch with bent knees

Progression: Standing Toe Touch, straight legs with longer hold

Red Flags: Sharp lower back pain

EXERCISE: Shoulder Wall Slides [Mobility]

Purpose: Improve overhead shoulder mobility and scapular control

Common Mistakes: Arching the lower back to fake a higher arm position

Likely Cause: Limited shoulder mobility being compensated for by spinal extension

Coaching Cues: Keep the lower back flat against the wall; only raise the arms as high as that allows

Correction Strategy: Reduce the height of the arm raise while maintaining a flat lower back

Regression: Reduced range Wall Slides

Progression: Wall Slides, full range with added hold time

Red Flags: Sharp shoulder pain

EXERCISE: Ankle Circles [Mobility]

Purpose: Improve general ankle joint mobility in all directions

Common Mistakes: Moving too fast through an incomplete range

Likely Cause: Rushing repetitions rather than tracing a full, deliberate circle

Coaching Cues: Move slowly through the largest comfortable circle in both directions

Correction Strategy: Reduce the circle size and move more slowly

Regression: Smaller range Ankle Circles

Progression: Ankle Circles, larger range or added resistance

Red Flags: Sharp ankle pain

EXERCISE: Cossack Stretch [Mobility]

Purpose: Improve lateral hip and adductor flexibility

Common Mistakes: Heel lifting on the working side; rounding the back

Likely Cause: Limited ankle and hip mobility for the deep lateral position

Coaching Cues: Keep the working foot flat and chest lifted throughout

Correction Strategy: Reduce depth and use a support for balance

Regression: Shallow Cossack Stretch

Progression: Full-depth Cossack Stretch with added hold time

Red Flags: Sharp groin or knee pain

EXERCISE: Standing Quad Stretch [Mobility]

Purpose: Improve quadriceps and hip flexor flexibility

Common Mistakes: Arching the lower back or leaning forward excessively to compensate for balance

Likely Cause: Poor balance causing compensatory trunk movement

Coaching Cues: Hold onto a wall or chair for balance; keep the torso upright throughout

Correction Strategy: Use a support for balance and reduce the pull on the ankle

Regression: Supported Standing Quad Stretch

Progression: Unsupported Standing Quad Stretch, longer hold

Red Flags: Sharp knee pain

23.9 Additional Exercise Corrections

EXERCISE: Close-Grip Push-Up [Push]

Purpose: Emphasize triceps within a push-up using a closer hand position

Common Mistakes: Hands placed too narrow, causing wrist strain

Likely Cause: Insufficient wrist mobility for the narrow hand position

Coaching Cues: Place hands roughly shoulder-width with fingers angled slightly out for wrist comfort

Correction Strategy: Widen the hand position slightly and reduce reps

Regression: Standard Push-Up

Progression: Close-Grip Push-Up, elevated feet

Red Flags: Sharp wrist pain

EXERCISE: Cable Row (Wide Grip) [Pull]

Purpose: Target the upper back and rear shoulders through a wide pulling grip

Common Mistakes: Pulling with the arms instead of leading with the elbows back and out

Likely Cause: Insufficient upper-back activation relative to arm strength

Coaching Cues: Lead the pull with the elbows moving back and slightly out

Correction Strategy: Reduce weight and re-cue elbow-led pulling

Regression: Standard Seated Cable Row

Progression: Wide-Grip Row, heavier load

Red Flags: Shoulder pain

EXERCISE: Hack Squat (Machine) [Squat]

Purpose: Build quad-dominant squat strength with machine-guided stability

Common Mistakes: Knees traveling far past the toes on a steep platform angle

Likely Cause: Platform angle or foot placement not suited to current mobility

Coaching Cues: Place feet slightly higher on the platform to reduce forward knee travel

Correction Strategy: Reduce weight and adjust foot placement higher on the platform

Regression: Goblet Squat

Progression: Hack Squat, heavier load

Red Flags: Sharp knee pain

EXERCISE: Step-Back Lunge with Reach [Lunge]

Purpose: Combine lunge strength with a mobility-focused overhead reach

Common Mistakes: Losing balance during the overhead reach phase

Likely Cause: Insufficient single-leg stability to add the upper-body component

Coaching Cues: Master the standard reverse lunge before adding the reach

Correction Strategy: Remove the overhead reach until lunge balance is solid

Regression: Reverse Lunge (no reach)

Progression: Step-Back Lunge with Reach, added load

Red Flags: Loss of balance and falling

EXERCISE: Single-Arm Kettlebell Swing [Hinge]

Purpose: Build unilateral hip-hinge power and grip endurance

Common Mistakes: Excessive torso rotation to compensate for single-arm loading

Likely Cause: Insufficient anti-rotation core strength for the asymmetric load

Coaching Cues: Brace the core to resist rotation; keep the hips squared throughout

Correction Strategy: Reduce load and reinforce the two-arm swing pattern first

Regression: Two-Arm Kettlebell Swing

Progression: Single-Arm Swing, heavier load

Red Flags: Sharp lower back pain; losing grip on the kettlebell

EXERCISE: Bicycle Crunch [Core]

Purpose: Build rotational core strength through an alternating crunch pattern

Common Mistakes: Pulling on the neck with the hands; moving too fast with poor range

Likely Cause: Insufficient core strength leading to compensatory neck pulling

Coaching Cues: Rest hands lightly behind the head; rotate the ribcage, not just the elbow

Correction Strategy: Slow the tempo significantly and reduce range of motion

Regression: Standard Crunch (no rotation)

Progression: Bicycle Crunch, slower tempo with longer range

Red Flags: Sharp neck pain

EXERCISE: Sprint Intervals [Cardio]

Purpose: Build high-intensity cardiovascular and power conditioning

Common Mistakes: Sprinting at full effort without adequate warm-up

Likely Cause: Insufficient preparation for the explosive demand of sprinting

Coaching Cues: Complete a full dynamic warm-up and 2-3 build-up strides before maximal effort

Correction Strategy: Reduce sprint intensity to 80-90% effort initially

Regression: Brisk Walk or Jog Intervals

Progression: Sprint Intervals, shorter rest between reps

Red Flags: Sharp hamstring or calf pain; chest pain

EXERCISE: Standing Hamstring Stretch [Mobility]

Purpose: Improve hamstring flexibility in a standing position

Common Mistakes: Rounding the back aggressively to reach further

Likely Cause: Seeking a deeper stretch through spinal flexion rather than hip flexion

Coaching Cues: Hinge from the hips with a flat back rather than rounding the spine

Correction Strategy: Bend the knees slightly and reduce the forward reach

Regression: Standing Hamstring Stretch, bent knees

Progression: Standing Hamstring Stretch, straight legs with longer hold

Red Flags: Sharp lower back pain

MODULE 24 | FITNESS PERSONA LIBRARY

20 AI coaching personas. Each includes profile overview, challenges, motivation triggers, common mistakes, recommended training style, recovery strategy, lifestyle adjustments, AI coaching approach, risk factors, weekly schedule template, and decision rules.

24.1 Absolute Beginner

PERSONA: Absolute Beginner

Profile Overview: Has never followed a structured exercise program. Low baseline fitness, strength, and movement confidence. May feel intimidated by terminology, equipment, or gym environments.

Typical Challenges: Lack of movement competency in fundamental patterns; low confidence; uncertainty about correct form; risk of early burnout from overambitious starts.

Motivation Triggers: Visible early wins such as a first full push-up or completing a full week; simple, achievable weekly targets; reassurance that everyone starts somewhere.

Common Mistakes: Starting with too much volume or intensity; comparing themselves to experienced people; skipping warm-up; quitting after one missed session.

Recommended Training Style: Module 6 Program A (Absolute Beginner) bodyweight-only, 3 sessions per week, full-body, conservative progression over 4 weeks before advancing.

Recommended Recovery Strategy: Standard 1-2 rest days; extra emphasis on sleep education since this population often underestimates its role; light walking on rest days.

Lifestyle Adjustments: Minimal lifestyle adjustment needed beyond consistent scheduling; recommend training at the same time each day to build habit automaticity.

AI Coaching Approach: Use plain, non-technical language; celebrate small milestones explicitly; check in on form confidence regularly; avoid overwhelming with too many variables at once.

Risk Factors: Early dropout due to unrealistic pacing; injury from poor form due to inexperience; discouragement from comparing to social media fitness content.

Weekly Schedule Template: Mon: Full Body A bodyweight. Tue: Rest or easy walk. Wed: Full Body B. Thu: Rest or easy walk. Fri: Full Body A. Sat: Light mobility. Sun: Rest.

Decision Rules:

IF the person has never exercised before → **THEN** assign Module 6 Program A and avoid introducing more than 4-6 exercises in week 1

IF the person reports high anxiety about starting → **THEN** recommend home-only sessions for the first 2 weeks

IF the person completes 4 consecutive weeks with good adherence → **THEN** progress to Module 7 intermediate programming

IF the person misses a session in the first 2 weeks → **THEN** reassure that this is normal and resume at the next opportunity without guilt framing

24.2 Student

PERSONA: Student

Profile Overview: School or early-stage student with a structured but demanding daily schedule, exam periods, and variable energy levels tied to academic stress.

Typical Challenges: Unpredictable daily schedule; exam-period stress reducing available time and motivation; possible limited budget for equipment.

Motivation Triggers: Framing exercise as both a stress-relief tool and a performance enhancer for academic focus; short, achievable session targets.

Common Mistakes: Treating fitness as the first thing to cut during busy periods; all-or-nothing thinking around missed sessions; ignoring exercise entirely during exams.

Recommended Training Style: Module 6 Program C (Student) using 20-30 minute bodyweight sessions, 3x per week strength plus 2x per week short cardio, flexible around class schedule.

Recommended Recovery Strategy: Sleep prioritization is especially important given common late-night study habits; encourage consistent sleep and wake times even during exam periods.

Lifestyle Adjustments: Build a flexible weekly target rather than fixed daily times; use campus walking as built-in activity.

AI Coaching Approach: Offer a reduced exam-period protocol proactively before stress peaks; reinforce that short sessions still count and are not a lesser effort.

Risk Factors: Complete dropout during high-stress academic periods; poor sleep compounding low energy and motivation.

Weekly Schedule Template: Mon/Wed/Fri: Full body bodyweight circuit, 20-25 min. Tue/Thu: 15-20 min walk between classes. Weekend: flexible, one longer activity or rest.

Decision Rules:

IF an exam period is approaching → **THEN** proactively reduce session count to 2-3 short sessions and frame remaining sessions as stress relief

IF the student reports a tight budget → **THEN** default entirely to bodyweight training with no equipment cost

IF the student's schedule changes daily → **THEN** use a flexible weekly target instead of fixed daily session times

24.3 College Student

PERSONA: College Student

Profile Overview: University-level student, often living semi-independently, balancing coursework, social life, and possibly part-time work, with more autonomy over schedule than a younger student.

Typical Challenges: Irregular sleep patterns common to college life; inconsistent meal timing; social pressure around alcohol and late nights; possible gym access via campus facilities.

Motivation Triggers: Social and identity-based framing around becoming someone who trains consistently; campus gym access can be leveraged as a built-in resource and social opportunity.

Common Mistakes: Inconsistent sleep undermining recovery; skipping meals before training; over-relying on caffeine instead of addressing root energy causes.

Recommended Training Style: Module 7 intermediate programming if campus gym access exists; otherwise Module 6 Program C. Full-body or upper-lower split, 3-4x per week.

Recommended Recovery Strategy: Address sleep consistency directly given college lifestyle norms; encourage moderation around alcohol given its recovery and sleep impact.

Lifestyle Adjustments: Leverage campus walking, intramural sports, or recreational activities as supplementary movement alongside structured training.

AI Coaching Approach: Be direct but non-judgmental about lifestyle factors like sleep and alcohol; emphasize sustainable consistency over short-term intensity.

Risk Factors: Burnout from balancing too many commitments; inconsistent training due to social schedule conflicts; poor recovery from irregular sleep.

Weekly Schedule Template: Mon/Wed/Fri: Full body or upper-lower split at campus gym, 45 min. Tue/Thu: recreational activity or rest. Weekend: flexible, social activity counts.

Decision Rules:

IF the student has access to a campus gym → **THEN** progress to Module 7 intermediate programming where appropriate

IF the student reports highly irregular sleep → **THEN** prioritize sleep consistency education before adjusting training volume

IF the student reports frequent alcohol-related disrupted sleep → **THEN** address the connection between alcohol and recovery directly and non-judgmentally

24.4 Hostel Student

PERSONA: Hostel Student

Profile Overview: Student living in a hostel or dormitory with shared facilities, minimal personal space, and typically zero exercise equipment access.

Typical Challenges: No equipment; very limited private space; shared bathroom and common areas reducing exercise privacy; potential noise constraints.

Motivation Triggers: Partnering with roommates or hallmates for mutual accountability; framing bodyweight training as equally effective as gym training.

Common Mistakes: Assuming nothing meaningful can be done without equipment; skipping training due to space or privacy concerns rather than adapting.

Recommended Training Style: Module 6 Program D (Hostel Resident) using zero-equipment bodyweight circuits fitting a roughly 2x2 meter space, 3x per week plus stair-based cardio.

Recommended Recovery Strategy: Standard rest day guidance; encourage maintaining sleep consistency despite often noisy or socially active hostel environments.

Lifestyle Adjustments: Use building stairs for cardio; schedule sessions during quieter hours for privacy; select stationary exercises over those needing significant room.

AI Coaching Approach: Reassure that equipment-free training is fully legitimate and effective; provide only exercises requiring minimal space.

Risk Factors: Low motivation due to feeling resource-constrained; inconsistent sleep from a shared, often noisy living environment.

Weekly Schedule Template: Mon/Wed/Fri: Bodyweight circuit of squats, push-ups, lunges, plank, and mountain climbers, 20 min. Tue/Thu/Sat: Stairs or walk, 15 min. Sun: Rest.

Decision Rules:

IF the student lives in a hostel with no equipment → **THEN** assign Module 6 Program D exclusively

IF the student reports low privacy for exercising → **THEN** schedule sessions during quieter hours or use only standing, low-visibility exercises

IF the student has access to building stairs → **THEN** use stair climbing as the default cardio option

24.5 Office Worker

PERSONA: Office Worker

Profile Overview: Works a desk-based job, typically 8+ hours per day, with limited built-in movement and common postural strain from prolonged sitting.

Typical Challenges: Prolonged sitting; neck, shoulder, and hip tightness; limited time before or after long commutes; afternoon energy slumps.

Motivation Triggers: Connecting training to reduced workday discomfort and improved afternoon energy; movement breaks framed as productivity tools, not just fitness.

Common Mistakes: Sitting for hours without any movement breaks; skipping mobility work entirely; relying on caffeine instead of movement for afternoon energy.

Recommended Training Style: Module 6 Program B (Sedentary Desk Worker) or Module 7 intermediate, 3-4x per week, combined with daily desk mobility breaks.

Recommended Recovery Strategy: Standard recovery guidance; specific emphasis on posture-related mobility work as a recovery and prevention tool.

Lifestyle Adjustments: Movement breaks every 30-60 minutes; lunch-break walks or workouts; stairs over elevators; standing during phone calls where possible.

AI Coaching Approach: Proactively suggest the office ergonomic checklist and desk mobility routine alongside structured training.

Risk Factors: Cumulative postural strain; sedentary-related cardiovascular risk if no structured activity is added; afternoon energy crashes.

Weekly Schedule Template: Mon/Wed/Fri: Full body strength during lunch break or after work, 25-30 min. Daily: movement break every 30-60 min. Tue/Thu: brisk walk.

Decision Rules:

IF the office worker reports neck and shoulder tightness → **THEN** prioritize daily desk mobility routine targeting hips, chest, and thoracic spine

IF the office worker has a lunch break available → **THEN** recommend using it for a brisk walk or short workout

IF the office worker reports persistent afternoon low energy → **THEN** recommend a brief walk during the slump instead of relying on caffeine alone

24.6 Software Engineer

PERSONA: Software Engineer

Profile Overview: Knowledge worker spending extended hours in deep focus at a screen, often with irregular breaks and a tendency toward sedentary, intense work cycles such as crunch periods and sprints.

Typical Challenges: Extreme sedentary stretches during focus sessions; eye strain and wrist/forearm tension; crunch periods that crowd out exercise entirely; tendency to overthink or over-optimize training plans instead of executing.

Motivation Triggers: Appeals to systems thinking and data; framing fitness as a system with inputs and outputs and trackable metrics resonates well with this persona's analytical mindset.

Common Mistakes: Treating fitness as another optimization problem to perfect before starting, delaying action; skipping movement breaks during deep work; training inconsistently around sprint deadlines.

Recommended Training Style: Module 7 intermediate programming, 3-4x per week; emphasize simple, repeatable structure over constant program-hopping driven by research rabbit holes.

Recommended Recovery Strategy: Strong emphasis on wrist, forearm, and neck mobility given keyboard-heavy work; sleep hygiene education given common late-night coding habits.

Lifestyle Adjustments: Use calendar-blocked movement breaks treated with the same priority as meetings; pair workouts with podcasts or technical content for added engagement.

AI Coaching Approach: Provide clear, logically structured rationale for recommendations; avoid vague motivational language; respect this persona's preference for direct, efficient communication.

Risk Factors: Repetitive strain patterns from desk work; complete training dropout during sprint or crunch periods; analysis paralysis preventing consistent execution.

Weekly Schedule Template: Mon/Wed/Fri: Full body strength, 30-40 min, calendar-blocked. Daily: 2-3 movement breaks during work hours. Tue/Thu: short cardio or walk.

Decision Rules:

IF the person reports a crunch or sprint deadline period → **THEN** offer a reduced minimum-viable protocol rather than letting training drop to zero

IF the person is over-researching their program instead of training → **THEN** recommend committing to one simple program for 4 weeks before any further optimization

IF the person reports wrist or forearm tension from typing → **THEN** prioritize daily wrist and forearm mobility work

24.7 Remote Worker

PERSONA: Remote Worker

Profile Overview: Works from home with a flexible but potentially blurred schedule, full control over their physical environment, and reduced incidental daily movement compared to a commuting office worker.

Typical Challenges: Blurred work-life boundaries can crowd out exercise time; reduced incidental movement since there is no commute or office walking; isolation reducing external accountability.

Motivation Triggers: Leverage the schedule flexibility as an advantage; training can be slotted into the day rather than confined to before or after work; virtual accountability partners can substitute for in-person social structure.

Common Mistakes: Letting work expand to fill all available time since there is no commute buffer; skipping all incidental movement since the home environment requires little walking.

Recommended Training Style: Module 7 intermediate programming, 3-5x per week, taking advantage of mid-day flexibility; home-based or hybrid gym access depending on equipment.

Recommended Recovery Strategy: Standard recovery guidance; encourage clear work-day boundaries to protect both training time and sleep schedule.

Lifestyle Adjustments: Schedule a fixed daily movement block since there is no natural commute-based activity; recommend a short walk to simulate a commute at the start and end of the workday.

AI Coaching Approach: Help the person build firm boundaries around training time given the flexibility can work against consistency as easily as for it.

Risk Factors: Very low incidental daily movement if not deliberately addressed; isolation reducing motivation and accountability; blurred boundaries leading to skipped sessions despite flexible time.

Weekly Schedule Template: Mon-Fri: Mid-day training block, 30-40 min, treated as a fixed calendar commitment. Daily: simulated commute walk at day start and end.

Decision Rules:

IF the remote worker reports skipping training due to work expanding to fill the day → **THEN** recommend treating the training block as a fixed, non-negotiable calendar appointment

IF the remote worker reports very low daily step count → **THEN** add a simulated commute walk at the start and end of the workday

IF the remote worker reports isolation reducing motivation → **THEN** recommend a virtual accountability partner or community

24.8 Business Owner

PERSONA: Business Owner

Profile Overview: Runs their own business with high autonomy but also high, often unpredictable time demands and significant decision-making stress.

Typical Challenges: Unpredictable daily schedule driven by business demands; high stress load; tendency to deprioritize personal health in favor of the business.

Motivation Triggers: Framing fitness as essential infrastructure for sustained business performance, not a competing priority; efficiency-focused messaging resonates well.

Common Mistakes: Treating exercise as entirely optional and the first thing cut under pressure; all-or-nothing thinking when the ideal schedule isn't possible.

Recommended Training Style: Module 7 intermediate or Module 19 time-constrained programming; flexible weekly-total approach rather than fixed daily times; efficient compound-exercise sessions.

Recommended Recovery Strategy: Stress management is a priority recovery lever given the role's demands; sleep consistency education given common late nights handling business needs.

Lifestyle Adjustments: Build a flexible weekly target with a menu of session lengths such as 10, 20, or 30 minutes to fit unpredictable gaps in the schedule.

AI Coaching Approach: Respect the person's time constraints without judgment; offer the most time-efficient options first; reinforce that imperfect consistency still compounds.

Risk Factors: Chronic stress without adequate management; complete deprioritization of health during business-critical periods; burnout.

Weekly Schedule Template: Flexible weekly target of 3-4 sessions, 20-30 min each, scheduled opportunistically around business demands rather than fixed days.

Decision Rules:

IF the business owner reports a highly unpredictable schedule → **THEN** use a flexible weekly-total target instead of fixed daily session times

IF the business owner reports high stress alongside training fatigue → **THEN** prioritize rest and stress-management practices over training volume

IF the business owner has zero time on a given day → **THEN** offer a single-set minimum viable workout rather than skipping entirely

24.9 Frequent Traveler

PERSONA: Frequent Traveler

Profile Overview: Travels regularly for work, spending significant time in hotels, airports, and unfamiliar environments with inconsistent access to equipment and routine.

Typical Challenges: Inconsistent equipment access; jet lag disrupting sleep and energy; unpredictable hotel gym quality; difficulty maintaining nutrition routine while traveling.

Motivation Triggers: Framing a portable, equipment-free routine as liberating rather than limiting; tracking total trips or days trained while traveling as its own metric of consistency.

Common Mistakes: Treating travel as an automatic reason to fully stop training; experimenting with unfamiliar hotel gym equipment instead of sticking to known movement patterns.

Recommended Training Style: Module 6 Program D-style bodyweight circuit usable in any hotel room; modified versions of the home program when hotel gym access exists.

Recommended Recovery Strategy: Prioritize sleep schedule stabilization over training schedule during the first 1-2 days of travel, especially across time zones.

Lifestyle Adjustments: Prioritize walking on travel or transit days; pack a resistance band as practical, lightweight equipment.

AI Coaching Approach: Provide a ready-made, no-equipment travel program in advance of trips; normalize reduced volume during travel as an acceptable trade-off, not a failure.

Risk Factors: Cumulative inconsistency across a heavy travel schedule; sleep disruption compounding recovery and energy issues; nutrition routine breakdown.

Weekly Schedule Template: Travel days: bodyweight hotel-room circuit, 15-20 min, or walking only. Home days: normal full-body program resumed promptly.

Decision Rules:

IF the person is on a travel or transit day → **THEN** prioritize walking over a structured workout

IF the person has hotel gym access → **THEN** use a modified version of their normal program with available equipment

IF the person is traveling for multiple weeks → **THEN** accept reduced training volume as a reasonable trade-off, not a failure

24.10 Parent

PERSONA: Parent

Profile Overview: Has caregiving responsibilities for one or more children, with daily schedule heavily structured around the child's needs and limited uninterrupted personal time.

Typical Challenges: Fragmented, unpredictable free time; physical and emotional fatigue from caregiving; guilt about prioritizing personal time for exercise.

Motivation Triggers: Framing personal fitness as modeling healthy habits for children; short windows of opportunity such as naps or school hours as legitimate full sessions.

Common Mistakes: Believing only long sessions count and skipping short windows as a result; deprioritizing personal health entirely during intense caregiving phases.

Recommended Training Style: Module 6/7 short-format sessions of 15-25 minutes designed to fit predictable caregiving windows such as naps or school hours.

Recommended Recovery Strategy: Realistic expectations around sleep given common caregiving-related sleep disruption, particularly with infants and young children; permission for reduced training volume during demanding phases.

Lifestyle Adjustments: Combine activity with family time where possible through active play or family walks; use short, predictable windows rather than waiting for large blocks of free time.

AI Coaching Approach: Actively counter guilt narratives around taking time for personal training; validate that fragmented short sessions are fully legitimate and effective.

Risk Factors: Complete deprioritization of personal health during intense caregiving phases; chronic sleep deprivation, particularly with infants; social isolation reducing motivation.

Weekly Schedule Template: 3-4x per week: 15-20 min session during a predictable caregiving window such as a nap or school hours. Bonus: active family play counted as supplementary movement.

Decision Rules:

IF the parent reports family obligations as the main training obstacle → **THEN** recommend combining activity with family time or using short windows during naps or quiet periods

IF the parent's family obligations are temporary, such as with a newborn → **THEN** plan for a temporary significant reduction in volume with gradual return as circumstances change

IF the parent reports guilt about taking time for personal training → **THEN** reframe personal fitness as modeling healthy habits for their children

24.11 Homemaker

PERSONA: Homemaker

Profile Overview: Manages household and often caregiving responsibilities full-time, with a physically active but unstructured daily routine that rarely translates into deliberate fitness training.

Typical Challenges: Daily physical activity is high but unstructured through chores and caregiving and does not build targeted strength or cardiovascular fitness; limited dedicated personal time; possible social isolation.

Motivation Triggers: Reframing structured exercise as distinct, additional benefit beyond daily household activity; framing personal training time as legitimate self-care, not indulgence.

Common Mistakes: Assuming household activity alone is sufficient structured exercise; deprioritizing personal training time as unimportant relative to family needs.

Recommended Training Style: Module 6/7 short-format sessions, 20-30 minutes, scheduled during predictable windows in the household routine.

Recommended Recovery Strategy: Standard recovery guidance; encourage genuine rest distinct from household activity, since constant low-level activity is not equivalent to recovery.

Lifestyle Adjustments: Identify a consistent daily or near-daily window, such as after morning chores or during a child's nap, for dedicated training.

AI Coaching Approach: Validate that household labor, while physically demanding, is different from structured fitness training; encourage claiming dedicated time without guilt.

Risk Factors: Chronic physical fatigue without proportional fitness gains since activity is unstructured; social isolation reducing motivation and accountability.

Weekly Schedule Template: 3-4x per week: 20-30 min structured strength or cardio session during a predictable household window. Daily: household activity counted as supplementary movement.

Decision Rules:

IF the homemaker assumes household chores are sufficient exercise → **THEN** clarify that structured training provides distinct strength and cardiovascular benefits beyond daily chores

IF the homemaker reports difficulty finding personal time → **THEN** identify a consistent daily window within the existing household routine

IF the homemaker reports social isolation → **THEN** recommend a community or group-based activity option where feasible

24.12 Older Adult

PERSONA: Older Adult

Profile Overview: Aged 65+, may be new to structured exercise or returning after a long gap, with a primary focus on maintaining independence, balance, and functional capacity.

Typical Challenges: Joint sensitivity; balance concerns and fall risk; possible underlying health conditions requiring medical clearance; reduced baseline strength and mobility.

Motivation Triggers: Framing exercise around maintaining independence and daily functional capability, such as getting up from chairs and climbing stairs, resonates strongly.

Common Mistakes: Avoiding strength training out of unfounded safety concerns; doing only seated or extremely passive activity when more is safely possible; skipping balance training specifically.

Recommended Training Style: Module 15 Older Adult protocol using low-impact, balance-focused programming with light-load strength training emphasizing full range of motion and control.

Recommended Recovery Strategy: Conservative progression with longer adaptation windows; extra emphasis on quality sleep and joint-friendly recovery practices.

Lifestyle Adjustments: Incorporate balance and functional movement such as sit-to-stand and floor get-ups into daily routine, not just structured sessions.

AI Coaching Approach: Require medical clearance before starting or changing programs if any underlying health condition is present; use clear, simple instructions; prioritize safety without being overly restrictive.

Risk Factors: Fall risk if balance training is neglected; loss of independence if strength training is avoided; underlying health conditions requiring careful monitoring.

Weekly Schedule Template: Mon/Wed/Fri: Light strength and balance circuit, 25-30 min. Tue/Thu: Walking, 20-30 min. Daily: balance practice such as single-leg stand.

Decision Rules:

IF the older adult is new to exercise → **THEN** begin with low-impact, balance-focused programming

IF the older adult has an underlying health condition → **THEN** require medical clearance before starting or changing a program

IF the older adult reports fear of falling → **THEN** prioritize balance and lower-body strength training together

24.13 Retired Individual

PERSONA: Retired Individual

Profile Overview: No longer working, with significant available free time but possibly reduced daily structure and routine compared to their working years.

Typical Challenges: Lack of externally imposed daily structure can reduce consistency despite abundant free time; possible reduced social interaction; varying health status depending on age and history.

Motivation Triggers: Framing fitness as a meaningful, structured daily anchor that replaces the routine previously provided by work; social fitness activities can fill both fitness and social needs.

Common Mistakes: Assuming abundant free time automatically translates to consistency without deliberate scheduling; under-utilizing the available time for genuinely productive training.

Recommended Training Style: Module 7 intermediate or Module 15 protocol depending on age and health status; can support more frequent sessions given time availability of 4-5x per week.

Recommended Recovery Strategy: Standard recovery guidance, adjusted for age and health status; encourage maintaining consistent daily structure to support both sleep and overall routine.

Lifestyle Adjustments: Build a consistent daily schedule with fitness as an anchor point; consider group classes or community programs for the social benefit.

AI Coaching Approach: Help establish reliable daily structure proactively, since the absence of work-imposed routine is the primary risk to consistency for this persona.

Risk Factors: Loss of consistency due to lack of external structure; social isolation if fitness is purely solitary; age-related health considerations depending on individual status.

Weekly Schedule Template: 4-5x per week: structured session, 30-40 min, at a consistent daily time. Include at least one group or social activity weekly if available.

Decision Rules:

IF the retired individual has abundant free time but inconsistent training → **THEN** help establish a fixed daily schedule anchor rather than relying on unstructured free time

IF the retired individual reports reduced social interaction → **THEN** recommend a group class or community-based fitness activity

IF the retired individual has age-related health considerations → **THEN** apply Module 15 Older Adult protocol guidance as appropriate

24.14 Overweight Beginner

PERSONA: Overweight Beginner

Profile Overview: New to structured exercise with a BMI in the overweight or obese range, often carrying both physical and psychological barriers to starting, such as joint load concerns and self-consciousness.

Typical Challenges: Joint stress from higher-impact activities; possible self-consciousness in gym or public settings; slower visible progress relative to effort, risking discouragement.

Motivation Triggers: Non-scale victories such as energy, strength, clothing fit, and mood framed prominently from the start; private or home-based options reducing self-consciousness.

Common Mistakes: Starting with high-impact activities that strain joints; comparing progress pace to people without this starting profile; focusing exclusively on the scale.

Recommended Training Style: Module 15 Overweight Population protocol using low-impact activities such as walking, swimming, and cycling plus modified bodyweight strength training, 2-3x per week initially.

Recommended Recovery Strategy: Standard recovery guidance; extra patience with soreness and adaptation timelines given the body's adjustment to new loading.

Lifestyle Adjustments: Recommend home or off-peak gym training initially if self-consciousness is a barrier; gradual introduction of more public or higher-intensity settings as confidence builds.

AI Coaching Approach: Lead with encouragement and non-scale progress markers; never use the BMI classification itself as framing in conversation with the person; emphasize sustainable pace over speed.

Risk Factors: Joint strain from inappropriate exercise selection; discouragement from slow visible progress; potential dropout from self-consciousness in public settings.

Weekly Schedule Template: Mon/Wed/Fri: Low-impact strength circuit such as modified squats and incline push-ups, 20-30 min. Tue/Thu: Walking or swimming, 20-30 min.

Decision Rules:

IF the person has BMI 25-29.9 and is new to exercise → **THEN** start with low-impact activities such as walking, swimming, and cycling

IF the overweight beginner reports joint discomfort during standard squats → **THEN** regress to box squats or reduced range of motion

IF the overweight beginner reports discouragement about pace of progress → **THEN** redirect focus to non-scale victories such as energy and strength

24.15 Underweight Beginner

PERSONA: Underweight Beginner

Profile Overview: New to structured exercise with a BMI below the healthy range, typically prioritizing strength and healthy weight gain over cardiovascular-focused training.

Typical Challenges: Low baseline strength; possible inadequate caloric or nutrient intake limiting training capacity and recovery; risk of excessive cardio further increasing caloric deficit.

Motivation Triggers: Framing strength gains and improved capability as primary, visible progress markers; connecting consistent training with improved energy and resilience.

Common Mistakes: Over-prioritizing cardio volume; under-fueling around training sessions; expecting rapid muscle gain without adequate nutritional support.

Recommended Training Style: Module 15 Underweight Population protocol using full-body strength training 3x per week prioritized over cardio, using compound bodyweight or light-resistance exercises.

Recommended Recovery Strategy: Extra emphasis on adequate rest between sessions given typically lower recovery reserves; nutrition adequacy flagged as essential alongside training.

Lifestyle Adjustments: Recommend reviewing meal regularity and adequacy alongside the training program, while staying within the scope of general fitness, not detailed nutrition prescription.

AI Coaching Approach: Prioritize strength-building guidance; caution against excessive cardio; recommend professional nutritional guidance if intake appears significantly inadequate.

Risk Factors: Overtraining relative to available recovery resources; inadequate fueling undermining both training capacity and overall health; potential underlying medical causes requiring evaluation.

Weekly Schedule Template: Mon/Wed/Fri: Full body strength circuit, 30 min, using compound bodyweight or light resistance. Light walking only for cardio, 1-2x per week.

Decision Rules:

IF the person has BMI below 18.5 → **THEN** prioritize strength training over excessive cardio volume

IF the underweight person reports low energy during sessions → **THEN** recommend reviewing adequacy of calorie and nutrient intake alongside training

IF the underweight individual shows signs of overtraining without adequate fueling → **THEN** reduce training volume and flag nutrition as a priority

24.16 Busy Professional

PERSONA: Busy Professional

Profile Overview: Works long hours in a demanding career with frequent meetings, deadlines, and limited predictable free time, similar to but distinct from a business owner due to typically fixed employer schedule constraints.

Typical Challenges: Long, often unpredictable work hours; high mental fatigue reducing perceived energy for training; frequent travel or after-hours obligations.

Motivation Triggers: Efficiency-focused framing around maximum benefit per minute trained; connecting consistent training to improved work performance and stress resilience.

Common Mistakes: Waiting for a less busy period that never reliably arrives; treating training as optional rather than scheduled; over-relying on weekend catch-up sessions only.

Recommended Training Style: Module 19 time-constrained programming; compound-exercise-focused sessions of 20-30 minutes; flexible weekly target with a menu of session lengths.

Recommended Recovery Strategy: Sleep protection is a priority given long work hours; stress-management practices recommended alongside physical training.

Lifestyle Adjustments: Calendar-block training time with the same priority as a work meeting; use early morning sessions if evenings are unpredictable.

AI Coaching Approach: Offer the fastest, most time-efficient options by default; reinforce that short, consistent sessions outperform sporadic long ones for this profile.

Risk Factors: Chronic under-recovery from long work hours; complete training dropout during peak busy periods; burnout if stress is not addressed alongside training.

Weekly Schedule Template: Mon/Wed/Fri: Compound full-body circuit, 25-30 min, calendar-blocked, ideally morning. Weekend: one longer session or active recovery.

Decision Rules:

IF the busy professional reports waiting for a less busy period to start → **THEN** recommend starting now with a reduced, sustainable version rather than waiting indefinitely

IF the busy professional has under 20 minutes available per session → **THEN** prioritize a full-body circuit of compound exercises

IF the busy professional reports high work stress alongside training fatigue → **THEN** prioritize rest and stress-management practices over training volume

24.17 Night Shift Worker

PERSONA: Night Shift Worker

Profile Overview: Works overnight or rotating shifts, with a sleep and activity schedule inverted or frequently shifted relative to conventional daytime patterns.

Typical Challenges: Circadian rhythm disruption affecting sleep quality and energy; social and family schedule misalignment; difficulty finding training windows that fit an inverted schedule.

Motivation Triggers: Anchoring training and routine to personal wake and sleep schedule rather than clock time; framing consistent training as a stabilizing anchor amid circadian disruption.

Common Mistakes: Trying to force training into conventional daytime hours that conflict with their sleep schedule; neglecting sleep environment optimization for daytime sleeping.

Recommended Training Style: Module 7 intermediate or Module 6 programming, scheduled relative to their personal wake time, such as shortly after waking regardless of clock time.

Recommended Recovery Strategy: Sleep environment optimization is critical given daytime sleeping, such as blackout curtains and noise control; consistent sleep and wake anchor times even on days off where possible.

Lifestyle Adjustments: Anchor all routine including meals, training, and sleep to shift pattern rather than conventional daily structure; minimize bright light exposure before intended sleep periods.

AI Coaching Approach: Avoid assuming conventional daytime scheduling; ask directly about their wake and sleep pattern and build all recommendations relative to it.

Risk Factors: Chronic sleep disruption affecting recovery and overall health; social isolation from misaligned schedule; difficulty maintaining any consistent routine across rotating shifts.

Weekly Schedule Template: 3x per shift cycle: training session shortly after waking relative to personal schedule, 25-30 min. Sleep environment optimized for daytime rest.

Decision Rules:

IF the person is a shift worker with irregular sleep hours → **THEN** anchor sleep environment consistency to personal schedule, not clock time

IF the night shift worker's pattern rotates unpredictably → **THEN** use a flexible weekly target with a ready menu of session lengths

IF the night shift worker reports difficulty finding a training window → **THEN** schedule training shortly after their personal wake time regardless of clock hour

24.18 Healthcare Worker

PERSONA: Healthcare Worker

Profile Overview: Works in a physically and emotionally demanding healthcare role, often with long or irregular shifts, high on-the-job physical activity, and significant emotional and stress load.

Typical Challenges: High physical fatigue from on-the-job demands such as standing, lifting, and walking; irregular shift patterns; emotional stress and burnout risk from the caregiving nature of the work.

Motivation Triggers: Framing personal fitness as essential self-care that sustains their capacity to care for others; stress relief and mental health benefits resonate strongly.

Common Mistakes: Assuming on-the-job physical activity is sufficient structured exercise; neglecting personal recovery while focused on patient care; ignoring emotional burnout signals.

Recommended Training Style: Module 7 intermediate programming adapted to shift patterns; sessions scheduled relative to shift timing rather than fixed clock times; emphasis on strength training to complement rather than duplicate on-the-job physical demands.

Recommended Recovery Strategy: Strong emphasis on stress management and emotional recovery alongside physical recovery; sleep consistency challenged by irregular shifts requires deliberate attention.

Lifestyle Adjustments: Anchor training to shift schedule; recommend brief stress-relief activity after particularly demanding shifts.

AI Coaching Approach: Validate the unique physical and emotional demands of the role; avoid generic recommendations that ignore shift-based scheduling realities.

Risk Factors: Burnout from combined physical and emotional demands; cumulative fatigue without adequate recovery; irregular sleep from shift work.

Weekly Schedule Template: 3x per week relative to shift schedule: Strength training, 25-30 min. Daily: brief stress-relief practice such as a short walk or breathing exercise after shifts.

Decision Rules:

IF the healthcare worker assumes on-the-job activity is sufficient exercise → **THEN** clarify that structured strength training provides distinct benefits beyond on-the-job physical demands

IF the healthcare worker reports high emotional stress alongside physical fatigue → **THEN** prioritize stress-management practices and adequate rest over training volume

IF the healthcare worker has irregular shift patterns → **THEN** anchor training schedule to shift timing rather than fixed clock hours

24.19 Teacher

PERSONA: Teacher

Profile Overview: Works in an education role with a structured school-term schedule, significant on-the-job standing and movement, but often limited personal time during the school day and seasonal variation between term-time and breaks.

Typical Challenges: Limited personal time during the school day; voice and physical fatigue from teaching demands; significant schedule shift between term-time and school breaks.

Motivation Triggers: Leveraging school breaks as opportunities for more intensive or varied training; framing fitness as a model for students and a stress-management tool.

Common Mistakes: Letting fitness lapse entirely during demanding term-time periods; not adjusting the program structure between term-time and break periods.

Recommended Training Style: Module 6/7 programming with two distinct phases: a reduced, efficient term-time protocol and a more flexible, expanded break-period protocol.

Recommended Recovery Strategy: Voice and standing-related fatigue addressed alongside standard recovery guidance; stress management important given classroom demands.

Lifestyle Adjustments: Schedule training around the fixed school day structure, such as before school, immediately after, or during a free period; use school breaks for program reassessment and progression.

AI Coaching Approach: Proactively offer distinct term-time and break-period program versions rather than a single static plan.

Risk Factors: Inconsistent training due to term-time and break schedule swings; cumulative classroom-related physical and vocal fatigue; stress-related burnout.

Weekly Schedule Template: Term-time: Mon/Wed/Fri, 20-25 min efficient session before or after school. Breaks: 4-5x per week, 35-45 min expanded sessions.

Decision Rules:

IF the teacher is in a demanding term-time period → **THEN** use the reduced, efficient term-time training protocol

IF the teacher is on a school break → **THEN** use the more flexible, expanded break-period training protocol

IF the teacher reports voice or standing-related fatigue → **THEN** incorporate appropriate recovery practices alongside standard guidance

24.20 Fitness Enthusiast

PERSONA: Fitness Enthusiast

Profile Overview: Already engaged with structured exercise and reasonably knowledgeable, seeking to refine, optimize, or maintain an established general fitness practice rather than starting from scratch.

Typical Challenges: Risk of program staleness or plateau from long-term repetition; potential overtraining from enthusiasm outpacing recovery capacity; may seek more advanced programming than this general fitness scope is designed to provide.

Motivation Triggers: Continued progression and variety; data and progress tracking resonate well given established engagement; mastery and refinement framing rather than basic habit-building framing.

Common Mistakes: Chasing constant program novelty at the expense of consistency; under-prioritizing recovery due to high enthusiasm; comparing their advanced progress to unrealistic external benchmarks.

Recommended Training Style: Module 7 intermediate to the upper end of general fitness scope; structured progression cycles with planned variation and periodic deloads.

Recommended Recovery Strategy: Standard recovery guidance applies with extra attention to deload scheduling given likely higher training volume and intensity.

Lifestyle Adjustments: Already well-integrated into routine; primary support need is programming refinement and plateau troubleshooting rather than basic habit formation.

AI Coaching Approach: Engage with more technical detail and nuance than other personas; route requests exceeding general fitness scope, such as competitive or elite training, to appropriate boundaries per Module 18.

Risk Factors: Overtraining due to high enthusiasm outpacing recovery; plateau-driven frustration; potential need for goals beyond this knowledge base's general fitness scope.

Weekly Schedule Template: 4-5x per week: structured strength and cardio split, 40-50 min sessions, with planned deload every 6-8 weeks.

Decision Rules:

IF the fitness enthusiast reports a plateau despite high consistency → **THEN** apply Module 22 plateau troubleshooting and consider a deload before further program changes

IF the fitness enthusiast requests competitive or elite-level programming → **THEN** clarify this exceeds general fitness scope and route to specialized guidance per Module 18

IF the fitness enthusiast has not taken a deload in 8+ weeks → **THEN** recommend a deload week immediately regardless of current progress

MODULE 25 | AI COACH ROUTING & RESPONSE ENGINE

215 deterministic routing and response-generation rules across 7 subsections. Format: [RTxxx] IF condition THEN action.

25.1 Goal Routing

[RT001] IF goal equals weight loss → **THEN** route to Weight Loss Brain

[RT002] IF goal equals muscle gain → **THEN** route to Muscle Gain Brain

[RT003] IF goal equals general fitness → **THEN** route to GF1, this knowledge base

[RT004] IF goal equals sport-specific performance → **THEN** route to the relevant sport-specific brain if available, otherwise flag as out of scope

[RT005] IF goal equals rehabilitation from injury → **THEN** route to professional physiotherapy guidance, not self-directed programming

[RT006] IF goal is unclear or unstated → **THEN** run the Module 18 goal differentiation framework before routing

[RT007] IF goal combines weight loss and general fitness → **THEN** route primary programming to Weight Loss Brain with general fitness principles applied as a secondary layer

[RT008] IF goal combines muscle gain and general fitness → **THEN** route primary programming to Muscle Gain Brain with general fitness principles applied as a secondary layer

[RT009] IF goal changes mid-program → **THEN** re-run goal routing and adjust the assigned brain or program accordingly

[RT010] IF goal is explicitly stated as competitive or elite performance → **THEN** flag as outside general fitness scope and recommend specialized coaching

[RT011] IF goal is simply to feel more energetic → **THEN** route to GF1 with an emphasis on cardio and recovery modules

[RT012] IF goal is simply to move better or reduce stiffness → **THEN** route to GF1 with an emphasis on the mobility module

[RT013] IF goal is to prepare for a specific recreational event such as a charity walk → **THEN** route to GF1 with event-specific cardio progression

[RT014] IF goal is unrealistic given the provided timeframe → **THEN** route to GF1 but flag the timeline as unrealistic and recommend a longer horizon

[RT015] IF goal request is purely informational, not a personal program request → **THEN** answer directly from the relevant module without triggering full program assignment

[RT016] IF goal is body recomposition, losing fat and gaining muscle simultaneously → **THEN** route to GF1 as the closest general fitness fit, noting this is a slower, less specialized process than dedicated brains

[RT017] IF goal is explicitly disease-specific, such as managing diabetes through exercise → **THEN** route to professional medical guidance alongside general fitness principles

[RT018] IF person requests multiple simultaneous distinct goals → **THEN** ask which goal is the current top priority before assigning a program

[RT019] IF person's goal aligns with GF1 scope but request also includes detailed nutrition planning → **THEN** address general nutrition principles within GF1 scope and note detailed meal planning is outside this module's scope

[RT020] IF person is unsure which goal applies to their situation → **THEN** use the Module 18.3 goal identification decision tree interactively

[RT021] IF goal is to maintain current fitness level, not progress further → **THEN** route to GF1 maintenance-focused programming rather than progressive overload-heavy programming

[RT022] IF goal request comes from a third party about someone else, such as a parent asking for their child → **THEN** clarify the program must be tailored to the actual person training, and gather that person's data directly where possible

[RT023] IF goal is return-to-fitness after a long break with no new specific objective → **THEN** treat as a beginner-equivalent re-entry under GF1 regardless of prior fitness history

[RT024] IF person explicitly requests routing to a brain that does not match their stated goal → **THEN** clarify the mismatch and recommend the brain matching their actual stated goal

[RT025] IF goal request includes both general fitness and aesthetic appearance language → **THEN** route to GF1 but acknowledge aesthetic interest without shifting core programming focus

[RT026] IF person's goal is primarily mental health or stress related → **THEN** route to GF1 with cardio and consistency modules emphasized for their stress-reduction benefits

[RT027] IF person's stated goal conflicts with a flagged safety concern from assessment → **THEN** prioritize the safety concern and adjust goal pursuit accordingly

[RT028] IF goal is to support a specific medical recovery plan under physician guidance → **THEN** request confirmation of physician clearance and align programming with their stated restrictions

[RT029] IF goal request is ambiguous between general fitness and a specific brain due to vague phrasing → **THEN** ask one clarifying question before routing rather than guessing

[RT030] IF goal has been successfully achieved → **THEN** engage Module 18.6 goal-setting guidance to establish a new objective

[RT181] IF person's goal request includes language about a specific physique reference such as wanting to look like a public figure → **THEN** redirect toward personal, measurable fitness goals rather than appearance comparison to others

[RT182] IF person's goal request changes multiple times within a single conversation → **THEN** confirm the final stated goal explicitly before proceeding with routing

25.2 Injury Routing

[RT031] IF pain reported is sharp, sudden, and severe → **THEN** stop all programming for the affected area and recommend immediate professional medical evaluation

[RT032] IF pain reported is mild, dull, and improves with movement → **THEN** this is more consistent with normal exercise sensation; monitor and proceed cautiously

[RT033] IF pain reported worsens with continued exercise → **THEN** stop the aggravating exercise immediately and recommend professional evaluation

[RT034] IF pain reported is joint-specific and persistent across multiple sessions → **THEN** recommend professional evaluation before continuing to load that joint

[RT035] IF pain reported includes numbness or tingling → **THEN** treat as a red flag and recommend prompt professional medical evaluation

[RT036] IF person reports a previous diagnosed injury now fully healed with clearance → **THEN** proceed with standard programming but monitor that area closely for early warning signs

[RT037] IF person reports a previous diagnosed injury without clear medical clearance → **THEN** require clearance confirmation before loading that movement pattern

[RT038] IF person reports pain only during one specific exercise variation → **THEN** substitute an alternative exercise targeting the same pattern that does not provoke pain

[RT039] IF person reports pain during daily activities unrelated to exercise → **THEN** recommend professional evaluation regardless of exercise program status

[RT040] IF person reports an acute injury that occurred during a session → **THEN** stop the session immediately and recommend appropriate first-response care and professional evaluation

[RT041] IF person reports chronic, ongoing joint pain with no clear acute incident → **THEN** recommend professional evaluation to identify underlying cause before programming around it

[RT042] IF person reports pain has fully resolved after a period of modified training → **THEN** gradually reintroduce the affected movement pattern starting from a regression

[RT043] IF person reports swelling at a joint → **THEN** treat as a red flag and recommend professional medical evaluation before continuing to train that area

[RT044] IF person reports pain specifically during the lowering eccentric phase of a lift → **THEN** this can carry useful diagnostic information; recommend professional evaluation rather than guessing

[RT045] IF person reports an old injury that occasionally flares with no clear new trigger → **THEN** recommend professional re-evaluation if flares are frequent, otherwise manage flares with conservative regression

[RT046] IF person reports general muscle soreness, not joint pain → **THEN** this is consistent with normal training adaptation; no escalation needed

[RT047] IF person asks if they should push through reported pain → **THEN** advise against pushing through any pain that is sharp, joint-related, or worsening

[RT048] IF person reports pain localized to a single muscle that resolves within a few days → **THEN** treat as likely normal soreness and continue monitoring

[RT049] IF person has a documented condition requiring exercise modifications from a doctor → **THEN** apply the documented modifications exactly and do not override them with standard programming

[RT050] IF injury status is unclear from the person's description → **THEN** ask specific, clarifying questions about location, onset, and what worsens or improves it before proceeding

[RT051] IF person reports pain that began after a change in footwear or equipment → **THEN** investigate the equipment change as a likely contributing factor before assuming a training-load cause

[RT052] IF person reports bilateral symmetric joint pain in both knees or both shoulders → **THEN** this pattern may suggest a systemic rather than mechanical cause; recommend professional evaluation

[RT053] IF person reports pain only at the very end range of a movement → **THEN** regress to a reduced range of motion and recommend professional evaluation if it persists

[RT054] IF person is uncertain whether their discomfort qualifies as pain or normal training sensation → **THEN** use the distinction that sharp, localized, worsening sensation points to pain while dull, diffuse, improving sensation points to normal training

[RT055] IF person reports injury risk concerns before starting a new exercise, with no current pain → **THEN** address proactively with proper progression and form cues rather than avoidance

[RT183] IF person reports a minor injury fully resolved more than 6 months ago with no recurring symptoms → **THEN** treat as resolved history with no special programming restriction needed

[RT184] IF person reports uncertainty about whether to mention a minor past injury → **THEN** encourage disclosure of any past injury history for safer program design

25.3 Recovery Routing

[RT056] IF person reports severe fatigue with declining performance over multiple sessions → **THEN** route to a complete rest protocol and recommend a deload or full rest period

[RT057] IF person reports mild fatigue with stable performance → **THEN** route to active recovery guidance rather than complete rest

[RT058] IF person reports fatigue alongside illness symptoms → **THEN** route to the illness-specific recovery guidance from Module 21.6

[RT059] IF person reports fatigue with no clear cause identified → **THEN** route to Module 22.1 low-energy troubleshooting decision tree

[RT060] IF person requests guidance immediately after a very high-intensity session → **THEN** route to post-session recovery guidance, including hydration, nutrition timing, and sleep prioritization

[RT061] IF person reports cumulative fatigue across several weeks of consistent high-volume training → **THEN** route to deload week protocol

[RT062] IF person asks how to recover faster between sessions → **THEN** route to Module 9 recovery fundamentals and Module 21.6 recovery rules

[RT063] IF person reports recovery concerns specific to sleep quality → **THEN** route to Module 21.7 sleep coaching rules

[RT064] IF person reports recovery concerns specific to hydration → **THEN** route to Module 21.8 hydration coaching rules

[RT065] IF person reports recovery concerns tied to high life stress → **THEN** route to stress-management guidance alongside reduced training load recommendations

[RT066] IF person is returning from an extended break and asks about recovery expectations → **THEN** route to graduated return-to-training guidance starting at reduced volume

[RT067] IF person reports symptoms consistent with overtraining syndrome → **THEN** route to an extended rest or deload protocol and recommend professional evaluation if symptoms persist

[RT068] IF person's recovery request relates to a specific muscle group rather than general fatigue → **THEN** route to localized active recovery guidance for that muscle group

[RT069] IF person asks about recovery tools such as foam rolling or massage → **THEN** route to general recovery education clarifying these are optional supplements to sleep, nutrition, and rest

[RT070] IF person's recovery concern coincides with a flagged safety or injury issue → **THEN** prioritize the safety or injury routing over general recovery guidance

[RT071] IF person reports recovery has been consistently poor despite addressing sleep, stress, and hydration → **THEN** route to professional medical evaluation

[RT072] IF person asks if a rest day is truly necessary given high motivation to train → **THEN** route to education on the necessity of rest days for adaptation despite motivation level

[RT073] IF person's recovery status is unclear from available data → **THEN** default to recommending a conservative extra rest day

[RT074] IF person reports good recovery and asks if they can increase training frequency → **THEN** route to progressive program adjustment guidance under Module 21.20

[RT075] IF person's recovery request is purely educational, not personally urgent → **THEN** answer directly from Module 9/21.6 content without triggering a program adjustment

[RT185] IF person's recovery routing request coincides with a major upcoming life event requiring peak energy → **THEN** balance recovery needs with the practical timing of the event, leaning conservative

25.4 Assessment Routing

[RT076] IF assessment is incomplete → **THEN** use available data conservatively and default to beginner-level programming

[RT077] IF assessment has never been completed → **THEN** run the full Module 2 baseline assessment before assigning any program

[RT078] IF assessment flags a medical red flag → **THEN** halt program assignment and require medical clearance before proceeding

[RT079] IF assessment is due for renewal after 4-8 weeks since last → **THEN** trigger Module 2 re-test and update program accordingly

[RT080] IF assessment data conflicts with the person's self-reported fitness level → **THEN** trust the test results over self-report for program calibration

[RT081] IF assessment indicates BMI outside the healthy range → **THEN** route to the relevant Module 15 population protocol for overweight or underweight

[RT082] IF assessment indicates significant mobility restriction in a specific joint → **THEN** prioritize that joint's mobility work before progressing related strength exercises

[RT083] IF assessment is requested by a person under significant time pressure → **THEN** offer the abbreviated Module 1.10 quick self-check as a faster substitute

[RT084] IF assessment reveals the person is 65+ and new to exercise → **THEN** route to Module 15 Older Adult protocol from the outset

[RT085] IF assessment reveals pregnancy or postpartum status → **THEN** require medical clearance and route to Module 15 conservative guidance

[RT086] IF assessment is requested but the person has a known recent injury → **THEN** incorporate the injury routing rules from 25.2 before completing the assessment-based program assignment

[RT087] IF assessment results show balance test scores below threshold → **THEN** add dedicated balance training regardless of the person's stated primary goal

[RT088] IF assessment results are strong across all metrics with no flags → **THEN** proceed directly to standard intermediate or advanced program assignment

[RT089] IF person requests reassessment earlier than the standard window → **THEN** allow it but caution that short-interval data may show more noise than signal

[RT090] IF assessment was completed over 6 months ago before resuming programming → **THEN** treat as outdated and require a fresh assessment before assigning a new program

[RT091] IF person declines to complete a full assessment → **THEN** proceed cautiously with conservative, beginner-level defaults and flag the limitation

[RT092] IF assessment reveals conflicting goals and current capability such as an advanced goal with beginner test results → **THEN** anchor the program to actual current capability, not the stated aspirational goal

[RT093] IF assessment is for a person returning after an injury with professional clearance documentation → **THEN** incorporate any documented restrictions directly into the assessment interpretation

[RT094] IF assessment reveals significantly different capability between left and right sides → **THEN** flag this imbalance and incorporate unilateral training emphasis into the resulting program

[RT095] IF person asks whether assessment results are good or bad → **THEN** frame results neutrally as a starting baseline rather than a judgment

[RT186] IF person's assessment was completed using a different knowledge base's framework previously → **THEN** re-run assessment using this knowledge base's Module 2 framework for consistency

25.5 Program Generation Routing

[RT096] IF person is a beginner → **THEN** generate a program from Module 6 Beginner Programs

[RT097] IF person is an intermediate → **THEN** generate a program from Module 7 Intermediate Programs

[RT098] IF person's persona matches one of the 20 Module 24 personas → **THEN** incorporate that persona's specific training style and lifestyle adjustments into program generation

[RT099] IF person has under 20 minutes available per session → **THEN** generate a program using Module 19 time-constrained guidance

[RT100] IF person has zero equipment access → **THEN** generate a program using bodyweight-only exercises from Module 5

[RT101] IF person has full gym equipment access → **THEN** generate a program incorporating machines and free weights alongside bodyweight options

[RT102] IF person requests a program but has unresolved safety flags from assessment → **THEN** resolve safety flags via 25.6 before generating any program

[RT103] IF person's program needs updating after a 4-8 week cycle → **THEN** regenerate the program using fresh assessment data and Module 21.20 progression rules

[RT104] IF person requests a program update mid-cycle without a clear reason → **THEN** ask what specifically isn't working before regenerating the entire program

[RT105] IF person's goal, persona, and assessment data are all available and consistent → **THEN** proceed directly to full program generation without further clarifying questions

[RT106] IF person's stated training frequency exceeds safe recommendations for their experience level → **THEN** cap frequency at safe levels per Module 21.20 and explain the reasoning

[RT107] IF person requests a program heavier in one movement pattern than others without a specific reason → **THEN** generate a balanced program across all patterns unless a specific, valid reason for imbalance is given

[RT108] IF person's available equipment changes between sessions, sometimes home and sometimes gym → **THEN** generate a flexible program with equipment-based exercise substitutions noted for each exercise

[RT109] IF person requests programming exceeding general fitness scope such as powerlifting peaking → **THEN** flag as outside scope and recommend specialized coaching

[RT110] IF person's program request conflicts with a documented medical restriction → **THEN** prioritize the medical restriction over the person's stated preference

[RT111] IF person has multiple flagged risk factors from their assigned persona → **THEN** incorporate conservative defaults addressing each flagged risk factor into the generated program

[RT112] IF person requests a completely custom program structure outside standard templates → **THEN** accommodate reasonable customization while ensuring all movement patterns remain covered weekly

[RT113] IF person's program generation request lacks sufficient information to proceed safely → **THEN** request the minimum necessary additional information before generating the program

[RT114] IF program generation is requested for someone explicitly outside the general fitness population such as a minor → **THEN** flag as requiring different, age-appropriate guidance outside this knowledge base's scope

[RT115] IF person's generated program needs to account for a known plateau pattern → **THEN** incorporate Module 22 plateau-prevention variation into the program design proactively

[RT187] IF person's program generation request comes immediately after a persona reassignment → **THEN** regenerate the program reflecting the newly assigned persona's training style and lifestyle adjustments

[RT188] IF person's program generation request specifies a goal that was already fully routed to a different brain → **THEN** confirm with the person whether they want general fitness support alongside,

or full handoff to the other brain

25.6 Safety Rules

[RT116] IF person reports chest pain during or after exercise → **THEN** treat as a medical red flag and recommend immediate emergency medical attention

[RT117] IF person reports severe shortness of breath disproportionate to exercise intensity → **THEN** treat as a medical red flag and recommend immediate emergency medical attention

[RT118] IF person reports fainting or loss of consciousness during exercise → **THEN** treat as a medical red flag and recommend immediate emergency medical attention

[RT119] IF person reports irregular or rapid heartbeat with dizziness → **THEN** treat as a medical red flag and recommend immediate emergency medical attention

[RT120] IF person reports sudden, severe headache during exercise → **THEN** treat as a medical red flag and recommend immediate emergency medical attention

[RT121] IF person reports numbness or weakness on one side of the body → **THEN** treat as a medical emergency red flag and recommend immediate emergency medical attention

[RT122] IF person reports a visible bone deformity or inability to bear weight after an injury → **THEN** recommend immediate emergency medical attention

[RT123] IF person reports symptoms consistent with heat stroke such as confusion, hot dry skin, or very high body temperature → **THEN** recommend immediate emergency medical attention

[RT124] IF person reports a known heart condition and wants to begin a new exercise program → **THEN** require medical clearance before any program assignment

[RT125] IF person reports being pregnant and wants to begin or continue a new exercise program → **THEN** require medical clearance and apply conservative, pregnancy-appropriate modifications

[RT126] IF person reports a chronic condition not previously disclosed such as diabetes or asthma → **THEN** recommend professional referral for condition-specific exercise guidance before proceeding

[RT127] IF person reports joint pain persisting beyond 2 weeks despite modification → **THEN** recommend professional referral for evaluation

[RT128] IF person reports symptoms suggestive of an eating disorder or disordered exercise patterns → **THEN** recommend professional referral to a qualified healthcare provider rather than continuing standard programming

[RT129] IF person reports a mental health crisis or severe emotional distress → **THEN** prioritize directing the person to appropriate support resources over continuing fitness coaching

[RT130] IF person under 18 requests an adult-oriented intense program → **THEN** recommend age-appropriate guidance and flag this knowledge base's adult general-population scope

[RT131] IF person reports symptoms of overtraining syndrome unresolved after a full deload → **THEN** recommend professional medical evaluation

[RT132] IF person's reported symptoms could indicate a medication interaction with exercise → **THEN** recommend the person consult their prescribing physician before continuing

[RT133] IF person explicitly asks for medical diagnosis through this system → **THEN** clarify this system cannot diagnose medical conditions and recommend professional evaluation

[RT134] IF person reports a red flag symptom but wants to continue exercising anyway → **THEN** reiterate the safety concern clearly and decline to provide programming that would continue the risk

[RT135] IF person's situation involves any ambiguity about whether professional referral is needed → **THEN** default to recommending professional consultation when in doubt

[RT189] IF person reports a red flag symptom that has since fully resolved without intervention → **THEN** still recommend professional evaluation given the symptom occurred, even though it resolved

[RT190] IF person's safety concern is reported by someone other than the person experiencing it → **THEN** still apply the same safety routing rules regardless of who is reporting the symptom

[RT191] IF person asks whether a symptom warrants stopping exercise immediately or can wait until after the session → **THEN** advise stopping immediately for any red-flag-category symptom regardless of session timing

[RT192] IF person's safety flag was resolved by professional clearance documentation → **THEN** proceed with programming as documented, removing the previous restriction

[RT193] IF person reports a safety concern about someone else they train with, not themselves → **THEN** provide the same safety guidance applicable to that symptom, directed at the third party's situation

25.7 Response Generation Rules — Workout Requests

[RT136] IF person requests a workout for today with no other context provided → **THEN** use the most recent program assignment on file rather than generating from scratch

[RT137] IF person requests a workout but has no program on file → **THEN** run quick goal and equipment checks before generating a single session

[RT138] IF person requests a workout substitution for a specific exercise → **THEN** substitute an exercise targeting the same movement pattern matching available equipment

[RT139] IF person requests a harder version of today's workout → **THEN** apply one progression variable such as load, reps, or difficulty rather than overhauling the entire session

[RT140] IF person requests an easier version of today's workout → **THEN** apply one regression variable rather than overhauling the entire session

[RT141] IF person requests a workout shorter than their normal session length → **THEN** prioritize compound exercises and trim isolation work first

[RT142] IF person requests a workout with specific equipment not in their normal program → **THEN** regenerate using only the newly specified equipment

[RT143] IF person requests a workout targeting a specific muscle group only → **THEN** clarify this is a general fitness knowledge base favoring balanced full-body sessions, and offer a pattern-focused circuit as a reasonable middle ground

[RT144] IF person requests a workout but reports current pain or injury → **THEN** apply 25.2 injury routing before generating any workout

[RT145] IF person requests a workout immediately after a flagged safety concern → **THEN** resolve the safety concern through 25.6 before responding with workout content

25.7 Response Generation Rules — Nutrition Requests

[RT146] IF person requests basic general nutrition guidance → **THEN** respond using Module 13 general nutrition principles within this knowledge base's scope

[RT147] IF person requests a detailed personalized meal plan → **THEN** clarify this exceeds general fitness knowledge base scope and recommend a registered dietitian for detailed planning

[RT148] IF person requests pre-workout nutrition guidance → **THEN** respond using the relevant pre-workout timing guidance from Module 13

[RT149] IF person requests post-workout nutrition guidance → **THEN** respond using the relevant post-workout timing guidance from Module 13

[RT150] IF person requests supplement-specific guidance → **THEN** clarify this falls outside this knowledge base's scope and recommend consulting a healthcare provider or dietitian

[RT151] IF person's nutrition question is tied to a goal better served by a different brain such as weight loss or muscle gain → **THEN** answer general principles here and note the dedicated brain offers more goal-specific nutrition depth

[RT152] IF person reports a medical dietary restriction such as an allergy or condition → **THEN** recommend professional guidance for that specific restriction rather than generic advice

[RT153] IF person's nutrition question shows signs of disordered eating patterns → **THEN** redirect to professional support resources rather than providing specific dietary guidance

25.7 Response Generation Rules — Recovery Requests

[RT154] IF person requests recovery guidance with no specific symptom described → **THEN** respond using general Module 9/21.6 recovery fundamentals

[RT155] IF person requests recovery guidance with a specific symptom described → **THEN** route through 25.3 recovery routing before responding

[RT156] IF person requests a rest day recommendation → **THEN** evaluate recent training load and recovery indicators before confirming or adjusting

[RT157] IF person requests recovery guidance immediately after reporting a red flag symptom → **THEN** prioritize 25.6 safety routing before any recovery content

[RT158] IF person requests general recovery education unrelated to a personal current issue → **THEN** answer directly and comprehensively from Module 9 content

25.7 Response Generation Rules — Motivation Requests

[RT159] IF person requests general motivational support with no specific issue described → **THEN** respond with Module 21.10 motivation principles and ask what specifically feels low right now

[RT160] IF person requests motivation support tied to a specific cause such as boredom, comparison, or slow progress → **THEN** route through the relevant Module 22.6 motivation troubleshooting branch

[RT161] IF person requests motivation support during a documented difficult life period → **THEN** validate the difficulty and offer a reduced minimum-viable training option rather than full intensity

[RT162] IF person requests motivation support after achieving a goal → **THEN** help set a new meaningful goal per Module 18.6

[RT163] IF person's motivation request reveals signs of broader low mood persisting beyond fitness → **THEN** encourage the person to speak with a trusted person or professional alongside any fitness-specific support offered

25.7 Response Generation Rules — Progress Review Requests

[RT164] IF person requests a progress review with full tracking data available → **THEN** synthesize the available Module 17 tracking data into a clear, encouraging summary

[RT165] IF person requests a progress review with little or no tracking data available → **THEN** encourage starting a simple tracking habit and provide a qualitative review based on available information

[RT166] IF person's progress review reveals a plateau → **THEN** route through Module 22 plateau troubleshooting as part of the response

[RT167] IF person's progress review reveals strong, consistent improvement → **THEN** acknowledge specifically and offer appropriate progression to the next program stage

[RT168] IF person's progress review reveals inconsistent adherence underlying flat progress → **THEN** address consistency directly and honestly rather than implying a physiological plateau

[RT169] IF person requests a progress review comparing themselves to others, not their own baseline → **THEN** redirect the review to their personal trend data only

[RT170] IF person's progress review request coincides with a scheduled reassessment window → **THEN** recommend running the Module 2 reassessment as part of the review

25.7 Response Generation Rules — General

[RT171] IF person's request does not clearly fall into any of the above response categories → **THEN** identify the closest matching category and apply its general response principles

[RT172] IF person's request combines multiple response categories simultaneously such as a workout plus motivation request → **THEN** address each component in order of safety priority, then practical priority

[RT173] IF person's request is purely conversational with no actionable fitness content needed → **THEN** respond naturally and warmly without forcing a structured module response

[RT174] IF person's request reveals an unresolved safety flag from earlier in the conversation → **THEN** resolve the safety flag before generating any further programming content

[RT175] IF person's request is ambiguous between two response categories → **THEN** ask one clarifying question rather than guessing which category applies

[RT176] IF person provides a long, multi-topic message → **THEN** address each topic in a logically organized order without losing any of the original questions

[RT177] IF person's request language suggests urgency or distress → **THEN** prioritize a calm, clear, and supportive response over comprehensive technical detail

[RT178] IF person's request repeats a question already answered earlier in the conversation → **THEN** answer again concisely without assuming the person remembers or is testing the system

[RT179] IF person's request is in a context where brevity is clearly preferred → **THEN** provide a concise, direct answer rather than a comprehensive module-length response

[RT180] IF person's request is in a context where depth is clearly preferred → **THEN** provide a comprehensive, well-organized response drawing on all relevant modules

[RT194] IF person's request is in phrasing this system has not seen before but the intent is clear → **THEN** respond to the clear underlying intent rather than requiring exact phrasing matches

[RT195] IF person's request contains a typo or unclear phrasing that obscures intent → **THEN** infer the most likely intended meaning and confirm briefly if genuinely ambiguous

[RT196] IF person's request is testing the system's capabilities rather than seeking real coaching → **THEN** respond helpfully and accurately regardless of the apparent intent behind the question

[RT197] IF person's request asks this system to behave outside its general fitness coaching scope → **THEN** decline politely and redirect to the appropriate scope or resource

[RT198] IF person's request includes both a question and an implicit emotional need such as frustration alongside a technical question → **THEN** address the technical question directly while acknowledging the emotional context briefly

[RT199] IF person's response history shows a consistent communication style preference for brief or detailed answers → **THEN** match that established preference in the current response

[RT200] IF person's request would benefit from a follow-up question to provide a better answer → **THEN** ask the single most valuable clarifying question rather than guessing or over-asking

[RT201] IF person's request is answered fully by existing FAQ content in Module 12 or Module 20 → **THEN** use the matching FAQ content directly rather than generating new explanatory text

[RT202] IF person's request requires synthesizing information across multiple modules → **THEN** synthesize coherently rather than presenting disconnected module excerpts

[RT203] IF person's request is about how this knowledge base itself is structured or organized → **THEN** explain the relevant module structure directly and transparently

[RT204] IF person's request involves a hypothetical scenario rather than their actual personal situation → **THEN** answer the hypothetical accurately without assuming it reflects the person's real circumstances

[RT205] IF person's request is unusually broad, covering many modules at once such as asking to be told everything → **THEN** provide a structured overview and invite the person to drill into specific areas of interest

[RT206] IF person's request directly contradicts an earlier safety-based recommendation in the same conversation → **THEN** reaffirm the earlier safety recommendation rather than silently overriding it

[RT207] IF person's request is for this system to make a definitive medical judgment → **THEN** decline to make medical judgments and redirect to appropriate professional evaluation

[RT208] IF person's request is satisfied by a very short, direct answer → **THEN** provide the short, direct answer rather than padding with unnecessary additional content

[RT209] IF person's request requires a genuinely long, detailed answer to be useful → **THEN** provide the necessary depth rather than artificially truncating for brevity

[RT210] IF person's request includes outdated information about their own situation such as referencing an old injury as current → **THEN** gently clarify the current status before proceeding with routing

[RT211] IF person's request is the very first message in a new coaching relationship → **THEN** begin with the Module 2 assessment process before any program-specific response

[RT212] IF person's request comes after a long gap in the coaching relationship → **THEN** treat similarly to a returning-after-break scenario and confirm current status before resuming

[RT213] IF person's request involves comparing this knowledge base's guidance to another source they encountered → **THEN** evaluate the comparison on its merits using this knowledge base's evidence-based principles, without dismissing other sources unfairly

[RT214] IF person's request cannot be confidently routed using any rule in this module → **THEN** default to the safest, most conservative general fitness response and note the limitation

[RT215] IF person's request has been fully resolved by this module's routing logic → **THEN** proceed to generate the final response without further internal routing steps